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|                          |                                |                                                                                                |              |                                                                              |          |  |
|                          | Requirement Tracability Matrix |                                                                                                |              |                                                                              |          |  |
| Feature                  | Requirement ID                 | Requirement Description                                                                        | Test case ID | Test case description                                                        | Status   |  |
| Functional Requirements  |                                |                                                                                                |              |                                                                              |          |  |
| Facility & Event Updates | FEU-01                         | The system shall display the current open/closed and availability status of facilities.        | TC_FEU_002   | Verify system shows current open/closed times correctly                      | Deferred |  |
|                          |                                |                                                                                                | TC_FEU_003   | Verify system shows availability status correctly                            | Passed   |  |
|                          | FEU-02                         | The system shall aggregate and display upcoming events with the event name, time and location. | TC_FEU_004   | Verify system display upcoming events with name, time and location correctly | Passed   |  |
|                          | FEU-03                         | The system shall display near real-time data.                                                  | TC_FEU_005   | Verify data updates without any delay                                        | Passed   |  |
|                          | FEU-04                         | The system shall display last retrieved data when live data cannot fetch.                      | TC_FEU_006   | Verify last saved data is shown when no internet                             | Passed   |  |
|                          |                                |                                                                                                |              |                                                                              |          |  |
| Transit & Navigations    | TN-01                          | The system shall display campus bus routes.                                                    | TC_TN_002    | Verify bus routes are are displayed correctly for each bus                   | Passed   |  |
|                          | TN-02                          | The system shall display arrival and departure time.                                           | TC_TN_003    | Verify arrival and departure times accurately displayed for each bus         | Passed   |  |
|                          | TN-03                          | The system shall provide updated schedules.                                                    | TC_TN_004    | Verify system shows latest transit shedule updates                           | Failed   |  |
|                          | TN-04                          | The system shall display current status such as on time, delayed or unavailable.               | TC_TN_005    | Verify bus status shows 'On Time' when no delay                              | Passed   |  |
|                          |                                |                                                                                                | TC_TN_006    | Verify bus status shows 'Delayed' when delay exists                          | Passed   |  |
|                          |                                |                                                                                                | TC_TN_007    | Verify bus status shows 'Unavailable' when no busses available               | Passed   |  |
|                          |                                |                                                                                                |              |                                                                              |          |  |

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|  | Club Hub                  | CH-01  | The system shall let students to discover and join active communities.                                                                      | TC_CH_002  | Verify student can join a club successfully                      | Failed      |
|  |                           | CH-02  | The system shall display active clubs, communities with details.                                                                            | TC_CH_001  | Verify all clubs are shown with correct details                  | Failed      |
|  |                           | CH-03  | The system shall feature student-led groups like coding club, cybersecurity club and etc.                                                   | TC_CH_003  | Verify student-led clubs available                               | Passed      |
|  |                           | CH-04  | The system shall display club activities including upcoming meetings and events.                                                            | TC_CH_004  | Verify system display club activities correctly                  | Deferred    |
|  |                           |        |                                                                                                                                             |            |                                                                  |             |
|  | Check-in & Rewards System | CRS-01 | The system shall provide a check-in button or action when user is at valid campus locations.                                                | TC_CRS_001 | Verify user can check-in at valid campus locations using QR scan | Blocked     |
|  |                           | CRS-02 | The system shall validate user location before awarding points                                                                              | TC_CRS_002 | Verify system validates location before awarding points          | Blocked     |
|  |                           | CRS-03 | Upon a successful validated check-in, the system shall award a defined number of engagement points (e.g. +10 points) to the user's profile. | TC_CRS_003 | Verify points are received after successful check-in             | Blocked     |
|  |                           | CRS-04 | The system shall prevent a student from checking in at the same location more than once within a defined cool down period.                  | TC_CRS_005 | Verify duplicate check-in is prevented                           | In Progress |
|  |                           | CRS-05 | Key locations for check-in include university library, cafeteria, club meetings, event venues and transport.                                | TC_CRS_004 | Verify check-in allowed only at valid campus locations           | In Progress |
|  |                           | CRS-06 | The system shall display students current points balance.                                                                                   | TC_CRS_006 | Verify current points balance displayed correctly                | Failed      |
|  |                           | CRS-07 | The system shall maintain and display a log of the student's past check-in events.                                                          | TC_CRS_007 | Verify check-in history is available and shown                   | Failed      |
|  |                           | CRS-08 | The system shall implement logic to allow points to be redeemed against defined rewards.                                                    | TC_CRS_008 | Verify rewards redemption works correctly                        | Deferred    |

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|  |                   |        |                                                                            |                |                                                        |          |
|  | User Login        | LOG-01 | Verify registered user can log in with valid Student ID and Password       | TC_LOG_STU_001 | Verify successful login with correct id and password   | Passed   |
|  |                   | LOG-02 | Verify user cannot log in with invalid credentials                         | TC_LOG_STU_014 | Login with incorrect id and password                   | Passed   |
|  |                   | LOG-03 | Verify user profile (including points balance) loads correctly after login | TC_LOG_STU_002 | Verify correct points balance displayed after login    | Failed   |
|  |                   | LOG-05 | Verify session expires after inactivity                                    | TC_LOG_STU_006 | Verify session expires after inactivity                | Deferred |
|  |                   | LOG-06 | Verify login page uses secure connection (HTTPS)                           | TC_LOG_STU_007 | Verify login uses HTTPS secure connection              | Deferred |
|  |                   | LOG-07 | Verify no sensitive data (password, tokens) appears in URL or logs         | TC_LOG_STU_008 | Verify no sensitive data in URL or logs after login    | Not Run  |
|  |                   | LOG-08 | Verify account lock after multiple failed login attempts                   | TC_LOG_STU_020 | Verify account lock after multiple failed attempts     | Deferred |
|  |                   | LOG-09 | Verify password is case sensitive                                          | TC_LOG_STU_004 | Verify password is case sensitive                      | Passed   |
|  |                   | LOG-10 | Verify password is masked during entry                                     | TC_LOG_STU_003 | Verify password is masked                              | Passed   |
|  |                   | LOG-11 | Verify SQL injection is prevented in login fields                          | TC_LOG_STU_016 | SQL injection in Student ID field                      | Passed   |
|  |                   |        |                                                                            | TC_LOG_STU_017 | SQL injection in Password field                        | Passed   |
|  |                   |        |                                                                            |                |                                                        |          |
|  | User Registration | REG-01 | Create a new student user profile with valid student ID and email          | TC_REG_001     | Verify student can register with all valid information | Failed   |
|  |                   | REG-02 | Attempt to create a duplicate profile (same student ID)                    | TC_REG_011     | Verify duplicate student id rejected                   | Blocked  |
|  |                   | REG-03 | Create profile with missing required fields                                | TC_REG_007     | Verify required field validation during registration   | Blocked  |
|  |                   | REG-04 | Create profile with invalid email format                                   | TC_REG_008     | Verify invalid email format rejected                   | Blocked  |

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|  |                                | REG-05 | After registration, verify data is encrypted/stored securely                                                                                             | TC_REG_005 | Verify password is stored encrypted                                                                | Failed  |
|  |                                | REG-06 | After registration, verify user starts with 0 engagement points                                                                                          | TC_REG_004 | Verify new student starts with 0 points                                                            | Blocked |
|  | Non-Functional Recuirements    |        |                                                                                                                                                          |            |                                                                                                    |         |
|  | User Profiles & Authentication | UPA-01 | The system must be able to create and manage student user profiles.                                                                                      | TC_UPA-001 | Verify profile creation after registration                                                         | Failed  |
|  |                                | UPA-02 | The system must implement measures to ensure data integrity (e.g points balances are accurate and cannot be easily manipulated).                         | TC_UPA-002 | Verify all data are accurate                                                                       | Failed  |
|  |                                |        |                                                                                                                                                          | TC_UPA-003 | Verify data cannot be manipulated                                                                  | Blocked |
|  |                                | UPA-03 | The system shall display the student's profile including their details                                                                                   | TC_UPA-004 | Verify student can view their profiles                                                             | Failed  |
|  |                                |        |                                                                                                                                                          |            |                                                                                                    |         |
|  | Data Management & Security     | DMS-01 | All student data, including profile information and reward points balances, must be stored securely.                                                     | TC_DMS_001 | Verify password stored securely (Weak storage cryptography)                                        | Passed  |
|  |                                | DMS-02 | Student IDs must be masked in all display contexts. The system must not expose sensitive student data in logs or UI and protect from unauthorized users. | TC_DMS_002 | Verify student ID is masked in all displayed content (Personally identifiable information leakage) | Failed  |
|  |                                |        |                                                                                                                                                          | TC_DMS_003 | Verify sensitive data is not exposed (Sensitive data leakage)                                      | Failed  |
|  |                                |        |                                                                                                                                                          | TC_DMS_004 | Verify system is protected from unauthorized users (Broken Access Control)                         | Blocked |
|  |                                | DMS-03 | The points balance must only be modifiable through server-side validated check-in logic, preventing client-side manipulation.                            | TC_DMS_005 | Verify points balance cannot be manipulated (Integrity violation)                                  | Blocked |
|  |                                |        |                                                                                                                                                          |            |                                                                                                    |         |

|  |                            |       |                                                                                                                           |           |                                                      |        |
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|  | Reliability & Availability | RA-01 | The system must handle all possible failures gracefully, without crashing or presenting unhandled exceptions to the user. | TC_RA_001 | Verify system handles database failure without crash | Passed |
|  |                            | RA-02 | In the event of a failed data feed, the application must display the most recently cached data.                           | TC_RA_002 | Verify recently cached data is shown                 | Passed |

| Role    | Test_ID        | Description                                                    | Test Category     | Inputs                                                                                                                             | Type     | Pre-conditions                          | Steps                                                                                                                         | Post-conditions                           | Expected results                                                      | Actual results                                                                         | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID     | Notes                                                                                                                          |
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| Student | TC_LOG_STU_001 | Verify successful student login with correct id and password   | Functional        | ID: STU_014<br>Password: dularik123pwd                                                                                             | Positive | Student must be registered              | 1. Open login page<br>2. Enter valid user Id<br>3. Enter valid password<br>4. Click login button                              | User session active                       | Student redirected to the dashboard                                   | Student successfully redirected to the dashboard                                       | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_002 | Verify correct points balance displayed after login            | Functional        | ID: STU_014<br>Password: dularik123pwd                                                                                             | Positive | Student has existing points             | 1. Login successfully<br>2. Observe points balance displayed                                                                  | User session active                       | Correct points balance is displayed                                   | Shows a static/hardcoded value (1450) as points balance for all users                  | Fail                    | Fail                       | Fail                       | Fail                       | BUG_LOG_01 | -                                                                                                                              |
|         | TC_LOG_STU_003 | Verify password is masked                                      | Security - Data   | Password: dularik123pwd                                                                                                            | Positive | Login page is open                      | 1.Type valid password in password field<br>2. Verify password shown as dots or asterisk                                       | Password remains masked                   | Password character's appear as dots or asterisk                       | Password remains masked properly                                                       | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_004 | Verify password is case sensitive                              | Functional        | ID: STU_014<br>Password: dularik123pwd/ Dularik123pwd/<br>DULARIK123pwd/ dularik123pwd                                             | Positive | Student registered with unique password | 1. Login with lowercase letters<br>2. Login with uppercase letters<br>3. Login with correct password<br>4. Click login button | No session created for incorrect attempts | Only matching case format succeeds                                    | Only matching case format succeeds                                                     | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_005 | Verify generic error appears when login failed                 | Security - Config | ID: wrong234<br>Password: Wrongpass123                                                                                             | Positive | Student may or may not exist            | 1. Login with wrong Id or password<br>2. Observe error message                                                                | User remains on the login page            | Generic error message displays without specifying the error           | Login failed but no generic error message appears                                      | Fail                    | Pass                       | Pass                       | Pass                       | BUG_LOG_02 | Cycle2: Issue fixed. Displays generic error message like "Invalid credentials. Try again!"                                     |
|         | TC_LOG_STU_006 | Verify session expires after inactivity                        | Security - Auth   | ID: STU_014<br>Password: dularik123pwd                                                                                             | Positive | Student is logged in                    | 1. Login successfully<br>2. Leave site for defined timeout<br>3. Try to access protected page                                 | Session invalidated                       | User redirected to login and requires re-authentication               | -                                                                                      | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -          | Cycle1: Session timeout for student not implemented<br>Postponed due to time limitation                                        |
|         | TC_LOG_STU_007 | Verify login uses HTTPS secure connection                      | Security - Config | URL: <a href="http://localhost:8081/live-campus-hub/api/v1/auth/login">http://localhost:8081/live-campus-hub/api/v1/auth/login</a> | Positive | Login page is accessible                | 1. Check browser URL bar<br>2. Inspect network traffic                                                                        | Page loaded securely                      | Application should use HTTPS protocol for all authentication requests | -                                                                                      | Not Applicable          | Not Applicable             | Not Applicable             | Not Applicable             | -          | Cycle1: Application is running on local environment using HTTP (localhost/127.0.0.1). HTTPS is not configured in current setup |
|         | TC_LOG_STU_008 | Verify no sensitive data in URL or logs after login            | Security - Data   | ID: STU_014<br>Password: dularik123pwd                                                                                             | Positive | Student logs in successfully            | 1. Check browser history after login<br>2. Check server logs                                                                  | Session active                            | No password or token appears in URL                                   | No sensitive data appears in URL or logs after login                                   | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_009 | Verify successful student login redirects to student dashboard | Authorization     | ID: STU_014<br>Password: dularik123pwd                                                                                             | Positive | Student logs in successfully            | 1. Verify student login redirects to student dashboard correctly                                                              | User session active                       | Student redirects to the student dashboard                            | Student redirects to the student dashboard correctly                                   | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_010 | Verify Id field is required                                    | Functional        | Student ID: empty<br>Password: dularik123pwd                                                                                       | Negative | Student is on the login page            | 1. Leave id field empty<br>2. Enter valid password<br>3. Click login button                                                   | No session created                        | Login denied. Error message appear                                    | Login denied. Displays "Please enter both your Student ID and Password." error message | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_011 | Verify password field is required                              | Functional        | Student ID: STU_014<br>Password: empty                                                                                             | Negative | Student is on the login page            | 1. Enter valid user Id<br>2. Leave password field empty<br>3. Click login button                                              | No session created                        | Login denied. Error message appear                                    | Login denied. Displays "Please enter both your Student ID and Password." error message | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_012 | Login with correct id and incorrect password                   | Functional        | Student ID: STU_014<br>Password: Wrongpass123                                                                                      | Negative | Student must be registered              | 1. Open login page<br>2. Enter valid user Id<br>3. Enter invalid password<br>4. Click login button                            | Login failed                              | User login denied and shows an error message                          | User login failed. But no error message displayed                                      | Fail                    | Pass                       | Pass                       | Pass                       | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again!"                                                                |
|         | TC_LOG_STU_013 | Login with incorrect id and correct password                   | Functional        | Student ID: wrong234<br>Password: dularik123pwd                                                                                    | Negative | Student must be registered              | 1. Open login page<br>2. Enter invalid user Id<br>3. Enter valid password<br>4. Click login button                            | Login failed                              | User login denied and shows an error message                          | User login failed. But no error message displayed                                      | Fail                    | Pass                       | Pass                       | Pass                       | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again!"                                                                |
|         | TC_LOG_STU_014 | Login with incorrect id and password                           | Functional        | ID: wrong234<br>Password: Wrongpass123                                                                                             | Negative | Student must be registered              | 1. Open login page<br>2. Enter invalid user Id<br>3. Enter invalid password<br>4. Click login button                          | Login failed                              | User login denied and shows an error message                          | User login failed. But no error message displayed                                      | Fail                    | Pass                       | Pass                       | Pass                       | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again!"                                                                |
|         | TC_LOG_STU_015 | Login with empty fields                                        | Functional        | Student ID: empty<br>Password: empty                                                                                               | Negative | Student must be registered              | 1. Open login page<br>2. Leave id empty<br>3. Leave password empty<br>4. Click login button                                   | Login failed                              | User login denied and shows an error message                          | User login failed. Error message appears                                               | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_016 | SQL injection in ID field                                      | Security - Input  | Student ID: ' OR '1'='1<br>Password: dularik123pwd                                                                                 | Negative | Login page loaded                       | 1. Open login page<br>2. Enter ' OR '1'='1 in id field<br>3. Enter normal password<br>4. Click login button                   | No session created                        | User login denied                                                     | User login failed                                                                      | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_017 | SQL injection in Password field                                | Security - Input  | Student ID: STU_014<br>Password: ' OR '1'='1                                                                                       | Negative | Login page loaded                       | 1. Open login page<br>2. Enter ' OR '1'='1 in password field<br>4. Click login button                                         | No session created                        | User login denied                                                     | User login failed                                                                      | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_018 | XSS in ID field                                                | Security - Input  | ID: <script>alert('XSS')</script><br>Password: dularik123pwd                                                                       | Negative | Login page is open                      | 1. Enter XSS payload in user id field<br>2. Enter any password<br>Click login button                                          | No alert popup                            | Script not executed                                                   | System did not crash. Script not executed. No visible error message displayed.         | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |
|         | TC_LOG_STU_019 | XSS in password field                                          | Security - Input  | ID: STU_014<br>Password: <img src=x onerror=alert(1)>                                                                              | Negative | Login page is open                      | 1. Enter valid user Id<br>2. Enter XSS payload in password field<br>Click login button                                        | No alert popup                            | Script not executed                                                   | System did not crash. Script not executed. No visible error message displayed.         | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                              |

|  |  |                |                                                                                                              |                   |                                                                                                                                    |          |                                                                                                                 |                                                                                                                                                                                                               |                                                                                                         |                                                                                                                                                   |                                                                                                                                                       |                 |                 |                |                |            |                                                                                                                                                                               |
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|  |  | TC_LOG_STU_020 | Brute force protection                                                                                       | Security - Auth   | ID: STU_014<br>Password: Wrongpass123 (5 times) then correct password dularik123pwd                                                | Negative | Student account exists                                                                                          | 1. Attempt to login 5 times with wrong password<br>2. Attempt 6th time with correct password                                                                                                                  | Account locked for 5-15 minutes                                                                         | Account locked after 5 fail attempts. 6th fails                                                                                                   | 6th attempt succeeds without account locking                                                                                                          | Fail            | Fail            | Fail           | Deferred       | BUG_LOG_03 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STU_021 | Very long password                                                                                           | Security - Input  | ID: STU_014<br>Password: A * 2000                                                                                                  | Negative | Login page is open                                                                                              | 1. Enter valid user Id<br>2. Enter 2000 character password<br>3. Click login button                                                                                                                           | No Server error or crash                                                                                | System rejects without crashing. Validate input correctly                                                                                         | User login failed without crashing. Missing validation                                                                                                | Fail            | Fail            | Fail           | Deferred       | BUG_LOG_04 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STU_022 | Verify sensitive student details (passwords) are not exposed and encrypted during login across network layer | Security - Auth   | ID: STU_014<br>Password: dularik123pwd                                                                                             | Negative | Login page is open. Developer tools is open on network with "Preserve log" enabled                              | 1. Enter valid student details<br>2. Click login button<br>3. Click login network request in dev tools<br>4. Check payload tab                                                                                | Login attempt recorded and unencrypted password remains visible in network payload tab                  | The password field value should be hashed or encrypted before transmission or the request must be strictly enforced over a secure HTTPS protocol. | The login credentials travel across the network in plain text (password: "dularik123pwd"). The application lacks HTTPS encryption or frontend hashing | Fail            | Fail            | Fail           | Deferred       | BUG_LOG_05 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STU_023 | Verify response parameters do not expose sensitive student data over an unencrypted connection               | Security - Auth   | ID: STU_014<br>Password: dularik123pwd                                                                                             | Negative | Login page is open. Developer tools is open on network with "Preserve log" enabled and application localstorage | 1. Enter valid student details<br>2. Click login button<br>3. Click login network request in dev tools<br>4. Check preview tab<br>5. Open application tab<br>6. Check localstorage details                    | Raw encoded JWT token, username and role expose in network preview tab or application localstorage      | JWT tokens, username and user roles must be securely encrypted or transmitted over secure HTTPS connection                                        | JWT token, username, and role is exposed in network tab and transmitted over an unencrypted HTTP connection                                           | Fail            | Fail            | Fail           | Deferred       | BUG_LOG_06 | Cycle2: Dev tools preview tab not displays any response data. But these sensitive data found in browser's localstorage in application<br><br>Postponed due to time limitation |
|  |  | TC_LOG_STU_024 | Verify backend rejects manipulated student login payloads                                                    | Security - Auth   | ID: STU_014<br>Password: dularik123pwd                                                                                             | Negative | Login page is open. Developer tools is open on console tab                                                      | 1. Copy successful login request as fetch command from network tab<br>2. Paste it in console tab<br>3. Change username to different username<br>4. Keep the password<br>5. Press enter and observe the result | Current user session stays active on dashboard. Request blocked and backend returns 403 Forbidden error | The server must validate details for specific Id and reject manipulated request with 401 Unauthorized or 403 Forbidden error                      | Backend successfully handles modified request and returns 403 Forbidden status                                                                        | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  |                |                                                                                                              |                   |                                                                                                                                    |          |                                                                                                                 |                                                                                                                                                                                                               |                                                                                                         |                                                                                                                                                   |                                                                                                                                                       |                 |                 |                |                |            |                                                                                                                                                                               |
|  |  |                |                                                                                                              |                   |                                                                                                                                    |          |                                                                                                                 |                                                                                                                                                                                                               |                                                                                                         |                                                                                                                                                   |                                                                                                                                                       |                 |                 |                |                |            |                                                                                                                                                                               |
|  |  | TC_LOG_STF_01  | Verify successful login with correct Id and password                                                         | Functional        | ID: STF_001<br>Password: hashen123#                                                                                                | Positive | User must be registered                                                                                         | 1. Open login page<br>2. Enter valid user Id<br>3. Enter valid password<br>4. Click login button                                                                                                              | Staff user session active                                                                               | User redirect to the dashboard                                                                                                                    | User successfully redirect to the dashboard                                                                                                           | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_02  | Verify password is masked                                                                                    | Security - Data   | Password: hashen123#                                                                                                               | Positive | Login page is open                                                                                              | 1.Type valid password in password field<br>2. Verify password shown as dots or asterisk                                                                                                                       | Password remains masked                                                                                 | Password characters appear as dots or asterisk.                                                                                                   | Password remains masked properly                                                                                                                      | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_03  | Verify password is case sensitive                                                                            | Functional        | ID: STF_001<br>Password: Hashen123#/<br>HASHEN123#/<br>hashen123#                                                                  | Positive | User registered with unique password                                                                            | 1. Login with lowercase letters<br>2. Login with uppercase letters<br>3. Login with correct password<br>4. Click login button                                                                                 | No session created for incorrect attempts                                                               | Only matching case format succeeds                                                                                                                | Only matching case format succeeds                                                                                                                    | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_04  | Verify generic error appears when login failed                                                               | Security - Config | ID: wrong234<br>Password: Wrongpass123                                                                                             | Positive | User may or may not exist                                                                                       | 1. Login with wrong Id or password<br>2. Observe error message                                                                                                                                                | User remains on the login page                                                                          | Generic error message displays without specifying the error                                                                                       | Login failed but no generic error message appears                                                                                                     | Fail            | Pass            | Pass           | Pass           | BUG_LOG_02 | Cycle2: Issue fixed. Displays generic error message like "Invalid credentials. Try again"                                                                                     |
|  |  | TC_LOG_STF_05  | Verify session expires after inactivity                                                                      | Security - Auth   | ID: STF_001<br>Password: hashen123#                                                                                                | Positive | User is logged in                                                                                               | 1. Login successfully<br>2. Leave idle for defined timeout<br>3. Try to access protected page                                                                                                                 | Session invalidated                                                                                     | User redirected to login and requires re-authentication                                                                                           | -                                                                                                                                                     | Not Implemented | Not Implemented | Deferred       | Deferred       | -          | Cycle1: Session timeout for staff not implemented<br><br>Postponed due to time limitation                                                                                     |
|  |  | TC_LOG_STF_06  | Verify login uses HTTPS secure connection                                                                    | Security - Config | URL: <a href="http://localhost:8081/live-campus-hub/api/v1/auth/login">http://localhost:8081/live-campus-hub/api/v1/auth/login</a> | Positive | Login page is accessible                                                                                        | 1. Check browser URL bar<br>2. Inspect network traffic                                                                                                                                                        | Page loaded securely                                                                                    | Application should use HTTPS protocol for all authentication requests                                                                             | -                                                                                                                                                     | Not Applicable  | Not Applicable  | Not Applicable | Not Applicable | -          | Cycle1: Application is running on local environment using HTTP (localhost:127.0.0.1). HTTPS is not configured in current setup                                                |
|  |  | TC_LOG_STF_07  | Verify no sensitive data in URL or logs after login                                                          | Security - Data   | ID: STF_001<br>Password: hashen123#                                                                                                | Positive | User logs in successfully                                                                                       | 1. Check browser history after login<br>2. Check server logs                                                                                                                                                  | Session active                                                                                          | No password or token appears in URL                                                                                                               | No sensitive data appears in URL or logs after login                                                                                                  | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_08  | Verify successful staff login redirects to admin dashboard                                                   | Authorization     | ID: STF_001<br>Password: hashen123#                                                                                                | Positive | User logs in successfully                                                                                       | 1. Verify staff login redirects to admin dashboard correctly                                                                                                                                                  | User session active                                                                                     | Staff redirects to the admin dashboard                                                                                                            | Staff also redirects to the student dashboard                                                                                                         | Fail            | Pass            | Pass           | Pass           | BUG_LOG_07 | Cycle1: Staff redirects to correct staff/admin dashboard                                                                                                                      |
|  |  | TC_LOG_STF_09  | Verify Id field is required                                                                                  | Functional        | ID: empty<br>Password: hashen123#                                                                                                  | Negative | User is on the login page                                                                                       | 1. Leave Id field empty<br>2. Enter valid password<br>3. Click login button                                                                                                                                   | No session created                                                                                      | Login denied. Error message appear                                                                                                                | Login denied. Displays "Please enter both your Student ID and Password." error message                                                                | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_10  | Verify password field is required                                                                            | Functional        | ID: STF_001<br>Password: empty                                                                                                     | Negative | User is on the login page                                                                                       | 1. Enter valid user Id<br>2. Leave password field empty<br>3. Click login button                                                                                                                              | No session created                                                                                      | Login denied. Error message appear                                                                                                                | Login denied. Displays "Please enter both your Student ID and Password." error message                                                                | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_11  | Login with correct Id and incorrect password                                                                 | Functional        | ID: STF_001<br>Password: Wrongpass123                                                                                              | Negative | User must be registered                                                                                         | 1. Open login page<br>2. Enter valid user Id<br>3. Enter invalid password<br>4. Click login button                                                                                                            | Login failed                                                                                            | User login denied and shows an error message                                                                                                      | User login failed. But no error message displayed                                                                                                     | Fail            | Pass            | Pass           | Pass           | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again"                                                                                                                |
|  |  | TC_LOG_STF_12  | Login with incorrect Id and correct password                                                                 | Functional        | ID: wrong234<br>Password: hashen123#                                                                                               | Negative | User must be registered                                                                                         | 1. Open login page<br>2. Enter invalid user Id<br>3. Enter valid password<br>4. Click login button                                                                                                            | Login failed                                                                                            | User login denied and shows an error message                                                                                                      | User login failed. But no error message displayed                                                                                                     | Fail            | Pass            | Pass           | Pass           | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again"                                                                                                                |
|  |  | TC_LOG_STF_13  | Login with incorrect Id and password                                                                         | Functional        | ID: wrong234<br>Password: Wrongpass123                                                                                             | Negative | User must be registered                                                                                         | 1. Open login page<br>2. Enter invalid user Id<br>3. Enter invalid password<br>4. Click login button                                                                                                          | Login failed                                                                                            | User login denied and shows an error message                                                                                                      | User login failed. But no error message displayed                                                                                                     | Fail            | Pass            | Pass           | Pass           | BUG_LOG_02 | Cycle2: Issue fixed. Displays "Invalid credentials. Try again"                                                                                                                |
|  |  | TC_LOG_STF_14  | Login with empty fields                                                                                      | Functional        | ID: empty<br>Password: empty                                                                                                       | Negative | User must be registered                                                                                         | 1. Open login page<br>2. Leave Id empty<br>3. Leave password empty<br>4. Click login button                                                                                                                   | Login failed                                                                                            | User login denied and shows an error message                                                                                                      | User login failed. Error message appears                                                                                                              | Pass            | Pass            | Pass           | Pass           | -          | -                                                                                                                                                                             |

|  |  |               |                                                                                                            |                  |                                                                                  |          |                                                                                                                 |                                                                                                                                                                                                               |                                                                                                         |                                                                                                                                                  |                                                                                                                                                      |      |      |      |          |            |                                                                                                                                                                               |
|--|--|---------------|------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | TC_LOG_STF_15 | SQL injection in ID field                                                                                  | Security - Input | ID: 'OR '1'='1<br>Password: hashen123#                                           | Negative | Login page loaded                                                                                               | 1. Open login page<br>2. Enter 'OR '1'='1 in Id field<br>3. Enter normal password<br>4. Click login button                                                                                                    | No session created                                                                                      | User login denied                                                                                                                                | User login failed                                                                                                                                    | Pass | Pass | Pass | Pass     | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_16 | SQL injection in Password field                                                                            | Security - Input | ID: STF_001<br>Password: 'OR '1'='1                                              | Negative | Login page loaded                                                                                               | 1. Open login page<br>2. Enter valid user Id<br>3. Enter 'OR '1'='1 in password field<br>4. Click login button                                                                                                | No session created                                                                                      | User login denied                                                                                                                                | User login failed                                                                                                                                    | Pass | Pass | Pass | Pass     | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_17 | XSS in ID field                                                                                            | Security - Input | ID: <script>alert('XSS')</script><br>Password: hashen123#                        | Negative | Login page is open                                                                                              | 1. Enter XSS payload in user Id field<br>2. Enter any password<br>Click login button                                                                                                                          | No alert popup                                                                                          | Script not executed                                                                                                                              | System did not crash. Script not executed. No visible error message displayed.                                                                       | Pass | Pass | Pass | Pass     | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_18 | XSS in password field                                                                                      | Security - Input | ID: STF_001<br>Password: <img src=x onerror=alert(1)>                            | Negative | Login page is open                                                                                              | 1. Enter valid user Id<br>2. Enter XSS payload in password field<br>Click login button                                                                                                                        | No alert popup                                                                                          | Script not executed                                                                                                                              | System did not crash. Script not executed. No visible error message displayed.                                                                       | Pass | Pass | Pass | Pass     | -          | -                                                                                                                                                                             |
|  |  | TC_LOG_STF_19 | Brute force protection                                                                                     | Security - Auth  | ID: STF_001<br>Password: Wrongpass123 (5 times) then correct password hashen123# | Negative | User account exists                                                                                             | 1. Attempt to login 5 times with wrong password<br>2. Attempt 6th time with correct password                                                                                                                  | Account locked for 5-15 minutes                                                                         | Account locked after 5 fail attempts, 6th fails                                                                                                  | 6th attempt succeeds without account locking                                                                                                         | Fail | Fail | Fail | Deferred | BUG_LOG_03 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STF_20 | Very long password                                                                                         | Security - Input | ID: STF_001<br>Password: A * 2000                                                | Negative | Login page is open                                                                                              | 1. Enter valid user Id<br>2. Enter 2000 character password<br>3. Click login button                                                                                                                           | No Server error or crash                                                                                | System rejects without crashing. Validate input correctly                                                                                        | User login failed without crashing. Missing validation                                                                                               | Fail | Fail | Fail | Deferred | BUG_LOG_04 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STF_21 | Verify sensitive staff details (passwords) are not exposed and encrypted during login across network layer | Security - Auth  | ID: STF_001<br>Password: hashen123#                                              | Negative | Login page is open. Developer tools is open on network with 'Preserve log' enabled                              | 1. Enter valid staff details<br>2. Click login button<br>3. Click login network request in dev tools<br>4. Check payload tab                                                                                  | Login attempt recorded and unencrypted password remains visible in network payload tab                  | The password field value should be hashed or encrypted before transmission or the request must be strictly enforced over a secure HTTPS protocol | The login credentials travel across the network in plain text (password: "dulark123pwd"). The application lacks HTTPS encryption or frontend hashing | Fail | Fail | Fail | Deferred | BUG_LOG_05 | Postponed due to time limitation                                                                                                                                              |
|  |  | TC_LOG_STF_22 | Verify response parameters do not expose sensitive staff data over an unencrypted connection               | Security - Auth  | ID: STF_001<br>Password: hashen123#                                              | Negative | Login page is open. Developer tools is open on network with 'Preserve log' enabled and application localStorage | 1. Enter valid staff details<br>2. Click login button<br>3. Click login network request in dev tools<br>4. Check preview tab<br>5. Open application tab<br>6. Check localStorage details                      | Raw encoded JWT token, username and role expose in network preview tab or application localStorage      | JWT tokens, username and user roles must be securely encrypted or transmitted over secure HTTPS connection                                       | JWT token, username, and role is exposed in network tab and transmitted over an unencrypted HTTP connection                                          | Fail | Fail | Fail | Deferred | BUG_LOG_06 | Cycle2: Dev tools preview tab not displays any response data. But these sensitive data found in browser's localStorage in application<br><br>Postponed due to time limitation |
|  |  | TC_LOG_STF_23 | Verify backend rejects manipulated staff login payloads                                                    | Security - Auth  | ID: STF_001<br>Password: hashen123#                                              | Negative | Login page is open. Developer tools is open on console tab                                                      | 1. Copy successful login request as fetch command from network tab<br>2. Paste it in console tab<br>3. Change username to different username<br>4. Keep the password<br>5. Press enter and observe the result | Current user session stays active on dashboard. Request blocked and backend returns 403 Forbidden error | The server must validate details for specific Id and reject manipulated request with 401 Unauthorized or 403 Forbidden error                     | Backend successfully handles modified request and returns 403 Forbidden status                                                                       | Pass | Pass | Pass | Pass     | -          | -                                                                                                                                                                             |



[illegible]

|  | Test_ID    | Description                                                         | Test Category    | Type     | Pre-conditions                                                   | Steps                                                                                                         | Post-conditions               | Expected results                                                         | Actual results                                                            | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID     | Notes                                                                                                       |
|--|------------|---------------------------------------------------------------------|------------------|----------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|------------|-------------------------------------------------------------------------------------------------------------|
|  | TC_FEU_001 | Verify system shows facility details correctly                      | Functional       | Positive | Valid facility details exist in database                         | 1. Navigate to the Facility updates section<br>2. View facility details                                       | -                             | Facility details displays correctly                                      | As for facility details system displays raw JSON values                   | Not Implemented         | Fail                       | Pass                       | Pass                       | BUG_FEU_01 | Cycle1: Facility & Event section not developed yet<br>Cycle3: Issues Fixed                                  |
|  | TC_FEU_002 | Verify system shows current open/closed times correctly             | Functional       | Positive | Valid open/closed times exist in database                        | 1. Navigate to the Facility updates section<br>2. Verify displayed open/closed times                          | -                             | System displays open/closed time correctly                               | -                                                                         | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -          | Postponed due to time limitation                                                                            |
|  | TC_FEU_003 | Verify system shows availability status correctly                   | Functional       | Positive | Avilability status exist in database                             | 1. Navigate to the Facility updates section<br>2. Verify displayed availability status                        | -                             | System displays statud correctly                                         | System displays "Quiet" , "Empty" statuses                                | Not Implemented         | Blocked                    | Pass                       | Pass                       | -          |                                                                                                             |
|  | TC_FEU_004 | Verify system display events with name, time and location correctly | Functional       | Positive | Gather correct names, times and location for every event         | 1. Open facility and event updates section<br>2. Verify every event displays correct event details            | -                             | Event details displays correctly for every event                         | System failed to fetch available events with details                      | Not Implemented         | Fail                       | Fail                       | Fail                       | BUG_FEU_02 | Cycle1: Facility & Event section not developed yet<br>Cycle2: Displays details correctly except event title |
|  | TC_FEU_005 | Verify data updates without any delay                               | Performance      | Positive | Internet connection available                                    | 1. Open facility and event updates section<br>2. Refresh page<br>3. Check if shown data changed               | -                             | Newest data appears immediately after refresh without any delay          | Cannot verify immediate update because page refresh causes redirect issue | Not Implemented         | Fail                       | Fail                       | Fail                       | BUG_FEU_03 | Cycle1: Facility & Event section not developed yet<br>Cycle2: Implemented but found a new bug               |
|  | TC_FEU_006 | Verify last saved data is shown when no internet                    | Functional       | Positive | Previous data exist in cache                                     | 1. Disable internet connection<br>2. Open facility and event updates section<br>3. View displayed information | Re-enable internet connection | Without displaying any error, previously cached data is shown            | Previously fetched data displays                                          | Not Implemented         | Not Implemented            | Pass                       | Pass                       | -          | Cycle1: Facility & Event section not developed yet<br>Cycle2: No caching mechanism implemented.             |
|  | TC_FEU_007 | Verify system displays 'Closed' outside operating hours             | Functional       | Positive | Gather operating hours for each facility                         | 1. Open facility and event updates section<br>2. Check status during and after operating hours                | -                             | Gym status shows 'Open' during and 'Closed' after operating hours        | -                                                                         | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -          | Postponed due to time limitation                                                                            |
|  | TC_FEU_008 | No duplicate events displayed                                       | Functional       | Positive | Data source send same event twice                                | 1. Navigate to Events section<br>2. Check for duplicate events                                                | -                             | Events appears only once in the list                                     | Database prevents duplicate entries                                       | Not Implemented         | Blocked                    | Pass                       | Pass                       | -          | Cycle1: Facility & Event section not developed yet                                                          |
|  | TC_FEU_009 | Data refreshes within expected time                                 | Performance      | Positive | Data refresh interval set to 30 seconds                          | 1. Change displayed data in database<br>2. Refresh the page<br>3. Observe facilities details again            | -                             | Facilities details update within the defined refresh interval            | Cannot verify page refresh causes redirect issue                          | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_FEU_03 | Cycle1: Facility & Event section not developed yet                                                          |
|  | TC_FEU_010 | Past events not displayed in upcoming events section                | Functional       | Positive | Gather few past events                                           | 1. Open facility and event updates section<br>2. View upcoming events list<br>3. Verify no past events shown  | -                             | No past events appear in upcoming events section                         | Past events appeared in upcoming events section                           | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_FEU_04 | Cycle1: Facility & Event section not developed yet                                                          |
|  | TC_FEU_011 | Verify system handles facility status with unexpected value         | Security - Input | Negative | Data source sends 'Unknown' or 'Maintenance' for facility status | 1. Open facility and event updates section<br>2. View facility status<br>3. Observe displayed value           | -                             | System displays original value or 'Status unavailable' without crashing  | System displays original value without crashing                           | Not Implemented         | Blocked                    | Pass                       | Pass                       | -          | Cycle1: Facility & Event section not developed yet                                                          |
|  | TC_FEU_012 | Verify system handles missing location for event                    | Security - Input | Negative | Data source sends other details without location for an event    | 1. Open facility and event updates section<br>2. View the event<br>3. Check if location available             | -                             | System displays 'Location not available' or empty field without crashing | System displays original value without crashing                           | Not Implemented         | Blocked                    | Pass                       | Pass                       | -          | Cycle1: Facility & Event section not developed yet                                                          |

| Test_ID   | Description                                                          | Test Category    | Type     | Pre-conditions                                          | Steps                                                                                                                                         | Post-conditions               | Expected results                                         | Actual results                                                                | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID    | Notes                                                                                                                |
|-----------|----------------------------------------------------------------------|------------------|----------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|-----------|----------------------------------------------------------------------------------------------------------------------|
| TC_TN_001 | Verify Transit & Navigation section displays valid details           | Functional       | Positive | User logged in                                          | 1. Login to the application<br>2. Navigate to Transit & Navigation section<br>3. Verify displayed data                                        | -                             | Bus details displays correctly according to the database | Failed to fetch database values                                               | Not Implemented         | Fail                       | Pass                       | Pass                       | BUG_TN_01 | Cycle 1: Data fetching not developed yet<br>Cycle2: Failed to fetch<br>Cycle3: Displays data correctly               |
| TC_TN_002 | Verify bus routes are displayed correctly for each bus               | Functional       | Positive | Gather current routes for each bus                      | 1. Open Transit & Navigation section<br>2. Check routes of all listed busses with source                                                      | -                             | Bus routes are displayed correctly for each bus          | Bus routes are displayed correctly for each bus                               | Not Implemented         | Blocked                    | Pass                       | Pass                       |           | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_003 | Verify arrival and departure times accurately displayed for each bus | Functional       | Positive | Gather arrival time and departure time for a chosen bus | 1. Open Transit & Navigation section<br>2. Find details for chosen bus<br>3. Check arrival and departure time with source                     | -                             | Arrival and Departure times display accurately           | Arrival and Departure times display accurately                                | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_004 | Verify system shows latest transit schedule updates                  | Functional       | Positive | Bus schedule updated 1 min ago                          | 1. Open Transit & Navigation section<br>2. Refresh page<br>3. Check if data matches latest schedule                                           | -                             | Most recent schedule data display                        | Cannot verify because refresh cause automatically redirection to profile page | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_TN_02 | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_005 | Verify bus status shows 'On Time' when no delay                      | Functional       | Positive | Chosen bus has no delay                                 | 1. Open Transit & Navigation section<br>2. Find chosen bus from list<br>3. Check status                                                       | -                             | Bus status shows 'On Time' correctly                     | Bus status shows 'On Time' when no delay                                      | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_006 | Verify bus status shows 'Delayed' when delay exists                  | Functional       | Positive | Chosen bus is delayed                                   | 1. Open Transit & Navigation section<br>2. Find chosen bus from list<br>3. Check status                                                       | -                             | Bus status shows 'Delayed' correctly                     | Bus status shows 'Delayed' when delay exist                                   | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_007 | Verify bus status shows 'Cancelled' when no busses available         | Functional       | Positive | Chosen bus is no available                              | 1. Open Transit & Navigation section<br>2. Find chosen bus from list<br>3. Check status                                                       | -                             | Bus status shows 'Unavailable' correctly                 | Displays "Cancelled" when busses unavailable                                  | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_008 | Verify last cached data shown when offline                           | Functional       | Positive | Previously bus data exists in cache                     | 1. Disable internet connection<br>2. Open Transit & Navigation section<br>3. View displayed bus information                                   | Re-enable internet connection | Last cached data is shown                                | -                                                                             | Not Implemented         | Not Implemented            | Not Implemented            | Not Implemented            | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing<br>Cycle3: No caching mechanism available |
| TC_TN_009 | Verify system handles missing arrival time for a bus smoothly        | Security - Input | Negative | Data source sends bus details without arrival time      | 1. Open Transit & Navigation section<br>2. View the bus details with missing arrival time<br>3. Check how arrival time field is displayed     | -                             | Arrival time field shows as empty without crashing       | Database added Not Null constraints to handle empty data properly             | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_010 | Verify system handles missing departure time for a bus smoothly      | Security - Input | Negative | Data source sends bus details without departure time    | 1. Open Transit & Navigation section<br>2. View the bus details with missing departure time<br>3. Check how departure time field is displayed | -                             | Departure time field shows as empty without crashing     | Database added Not Null constraints to handle empty data properly             | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |
| TC_TN_011 | Verify system handles missing bus route smoothly                     | Security - Input | Negative | Data source sends bus details without route             | 1. Open Transit & Navigation section<br>2. View the bus details without route<br>3. Check how route is displayed                              | -                             | System displays 'Route unavailable' without crashing     | Database added Not Null constraints to handle empty data properly             | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing                                           |

|  |           |                                                            |                  |          |                                                        |                                                                                                                     |   |                                                                |                                                                   |                 |         |      |      |   |                                                                            |
|--|-----------|------------------------------------------------------------|------------------|----------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------|-------------------------------------------------------------------|-----------------|---------|------|------|---|----------------------------------------------------------------------------|
|  | TC_TN_012 | Verify system handles invalid time format without crashing | Security - Input | Negative | Data source sends bus details with invalid time format | 1. Open Transit & Navigation section<br>2. View the bus details with invalid time<br>3. Check how time is displayed | - | System displays 'Time not available' or empty without crashing | Database automatically adjust time according to the data type     | Not Implemented | Blocked | Pass | Pass | - | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing |
|  | TC_TN_013 | Verify system handles empty data source without crashing   | Security - Input | Negative | Data source sends empty details                        | 1. Open Transit & Navigation section<br>2. Observe displayed content                                                | - | Without crashing, system displays 'Bus not Available'          | Database added Not Null constraints to handle empty data properly | Not Implemented | Blocked | Pass | Pass | - | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing |
|  | TC_TN_014 | Verify system handles duplicate bus entries                | Functional       | Negative | Data source sends same bus twice                       | 1. Open Transit & Navigation section<br>2. Check for any duplicate busses                                           | - | Duplicate may appear but app does not crash                    | Database prevents duplicate entries                               | Not Implemented | Blocked | Pass | Pass | - | Cycle1: Data fetching not developed yet<br>Cycle2: Cannot continue testing |

|  | Test_ID   | Description                                                   | Test Category             | Type     | Pre-conditions                                                                                  | Steps                                                                                                                            | Post-conditions                             | Expected results                                                                                      | Actual results                                                                                                                                                | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID    | Notes                                                                                                                             |
|--|-----------|---------------------------------------------------------------|---------------------------|----------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
|  | TC_CH_001 | Verify all clubs are shown with correct details               | Functional                | Positive | Gather each available clubs with details                                                        | 1. Navigate to Club Hub section<br>2. View all clubs<br>3. Check each club's details                                             | -                                           | All clubs are displayed with correct information                                                      | Failed to fetch and display club data                                                                                                                         | Not Implemented         | Fail                       | Fail                       | Fail                       | BUG_CH_01 | Cycle1: Club Hub section not developed yet                                                                                        |
|  | TC_CH_002 | Verify student can join a club successfully                   | Functional                | Positive | Student is logged in. Student is not a member of chosen club                                    | 1. Navigate to Club Hub section<br>2. Find chosen club<br>3. Click 'Join' button<br>4. Confirm join action                       | Student is added as a member of chosen club | Success feedback appear 'Joined [club name] successfully'. Club appears in 'My Clubs' section         | Join works and data stores correctly in database but failed to display joined club's name, joined date. Also system displays other students' joined clubs too | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_02 | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_003 | Verify student-led clubs available                            | Functional                | Positive | System has available clubs                                                                      | 1.Open Club Hub section<br>2. Check which clubs appear                                                                           | -                                           | Clubs section includes student-led clubs                                                              | Clubs section includes student-led clubs                                                                                                                      | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Club Hub section not developed yet                                                                                        |
|  | TC_CH_004 | Verify system display club activities correctly               | Functional                | Positive | Chosen club has an activity                                                                     | 1. Navigate to Club Hub section<br>2. Find chosen club<br>3. View activities list                                                | -                                           | Activity is visible with name, time and location correctly                                            | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |
|  | TC_CH_005 | Verify student can leave a club successfully                  | Functional                | Positive | Student is logged in. Student is currently a member of chosen club                              | 1. Navigate to Club Hub section<br>2. Go to 'My Clubs' or chosen club<br>3. Click 'Leave Club' button<br>4. Confirm leave action | Student is removed from chosen club         | Success feedback appear 'Left [club name] successfully'. Club no longer appears in 'My Clubs' section | Uses wrong HTTP method and caused and error                                                                                                                   | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_03 | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_006 | Verify student cannot join same club twice                    | Functional                | Positive | Student is logged in. Student is already a member of chosen club                                | 1. Navigate to Club Hub section<br>2. Find chosen club<br>3. Check if 'Join' button still visible and clickable                  | -                                           | Join' button is disabled or shows 'Already Joined'                                                    | Duplicate entries prevented but displays success message                                                                                                      | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_04 | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_007 | Verify student can view clubs they've joined                  | Functional                | Positive | Student is logged in. Student is already joined to few clubs                                    | 1. Navigate to Club Hub section<br>2. Find 'My Clubs' or similar section<br>3. View list of joined clubs                         | -                                           | Already joined clubs appear in 'My Clubs' section                                                     | Joined club appears with other students' joined clubs                                                                                                         | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_02 | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_008 | Verify each club include member count                         | Functional                | Positive | Chosen club has 30 members                                                                      | 1. Navigate to Club Hub section<br>2. Find chosen club<br>3. Check if member count visible                                       | -                                           | Chosen club shows member count as '30 Members'                                                        | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |
|  | TC_CH_009 | Verify system handles when no clubs available                 | Functional                | Positive | No clubs are available in the system                                                            | 1. Navigate to Club Hub section<br>2. Observe displayed content                                                                  | -                                           | System displays 'No clubs available at this time' or similar                                          | Failed to notify users about clubs availability                                                                                                               | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_05 | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_010 | Verify system handles join attempt for non-existent club      | Security - Input          | Negative | Student is logged in. chosen club does not exist                                                | 1. Attempt to join club with invalid id or via URL manipulation or API<br>2. Observe system response                             | -                                           | System displays 'Club not found'                                                                      | -                                                                                                                                                             | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -         | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data<br>Cycle3: Cannot manipulate url with a static html ur |
|  | TC_CH_011 | Verify system handles missing club description smoothly       | Security - Input          | Negative | Data source sends club details without description                                              | 1. Navigate to Club Hub section<br>2. View the club with missing description<br>3. Check description field                       | -                                           | Description field shows 'No description available' or empty without crashing                          | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |
|  | TC_CH_012 | Verify system handles missing club meeting time smoothly      | Security - Input          | Negative | Data source sends club details without including meeting time                                   | 1. Navigate to Club Hub section<br>2. View the club with missing time<br>3. Check meeting time field                             | -                                           | Meeting time field shows 'Time not available' or empty without crashing                               | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |
|  | TC_CH_013 | Verify system handles duplicate clubs smoothly                | Functional                | Negative | Data source sends a club twice                                                                  | 1. Navigate to Club Hub section<br>2. Find all available clubs<br>3. Check for duplicate clubs                                   | -                                           | Both may appear but app does not crash                                                                | Database prevents duplicate entries                                                                                                                           | Not Implemented         | Blocked                    | Pass                       | Pass                       | -         | Cycle1: Club Hub section not developed yet<br>Cycle2: Failed to fetch data                                                        |
|  | TC_CH_014 | Verify system handles join club when club is at full capacity | Security - Business logic | Negative | Student is logged in. Chosen club has maximum capacity of 10 members and already has 10 members | 1. Navigate to Club Hub section<br>2. Find chosen club<br>3. Click 'Join' button                                                 | Student is not added as member              | Error appear 'Club is full'                                                                           | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |
|  | TC_CH_015 | Verify system handles clubs with missing location             | Security - Input          | Negative | Data source sends club details without including location                                       | 1. Navigate to Club Hub section<br>2. View the club with missing location<br>3. Check location field                             | -                                           | Location field shows 'Location not available' or empty without crashing                               | -                                                                                                                                                             | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -         | Postponed due to time limitation                                                                                                  |

| Test_ID    | Description                                                      | Test Category             | Type     | Pre-conditions                                                                             | Steps                                                                                                          | Post-conditions                                          | Expected results                                                               | Actual results                                                            | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID     | Notes                                                                                   |
|------------|------------------------------------------------------------------|---------------------------|----------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|------------|-----------------------------------------------------------------------------------------|
| TC_CRS_001 | Verify user can check-in at valid campus locations using QR scan | Functional                | Positive | Student is logged in. Student is at chosen valid location. QR code is available            | 1. Open check-in section<br>2. Open QR scanner<br>3. Scan QR in chosen location<br>4. Confirm check-in         | Check-in record is created                               | System displays 'Checked in at [location name]'. Points awarded                | -                                                                         | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented<br><br>Cycle3: Button not working               |
| TC_CRS_002 | Verify system validates location before awarding points          | Security - Business logic | Positive | Student is logged in. Student scans QR code                                                | 1. Scan QR code<br>2. System checks if location is valid<br>3. Observe points awarding                         | -                                                        | System validates the location and award points                                 | -                                                                         | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_003 | Verify points are received after successful check-in             | Functional                | Positive | Student is logged in. Student has 10 points currently. Check-in awards 5 points            | 1. Open check-in page<br>2. Perform check-in at a valid location<br>3. Check points balance after check-in     | Points balance updated                                   | Points balance increases to 15                                                 | -                                                                         | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_004 | Verify check-in allowed only at valid campus locations           | Security - Business logic | Positive | Student is logged in                                                                       | 1.Try to check-in at valid locations<br>2.Try to check-in at invalid location                                  | -                                                        | Check-in successful at valid locations. Check-in rejected at invalid locations | -                                                                         | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_005 | Verify duplicate check-in is prevented                           | Security - Business logic | Positive | Student is logged in. Student already checked in at chosen valid location                  | 1. Try to check-in at chosen location again in same day<br>2. Scan QR code again<br>3. Observe system response | No second check-in recorded. No duplicate points awarded | Error appears 'You've already checked in here today'                           | -                                                                         | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_006 | Verify current points balance displayed correctly                | Functional                | Positive | Student has 150 points                                                                     | 1. Log in to the application<br>2. Navigate to profile or rewards section<br>3. View points balance            | -                                                        | Points balance shows '150' correctly                                           | Failed to display student's points balance. Displays earned reward points | Not Implemented         | Not Implemented            | Fail                       | Fail                       | BUG_CRS_01 | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_007 | Verify check-in history is available and shown                   | Functional                | Positive | Student has checked in at few valid locations                                              | 1. Navigate to Check-in history<br>2. View list of past check-ins<br>3. Verify dates and locations             | -                                                        | All check-ins appear with correct dates and locations                          | System displays all students' check-in history                            | Not Implemented         | Not Implemented            | Fail                       | Fail                       | BUG_CRS_02 | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_008 | Verify rewards redemption works correctly                        | Functional                | Positive | Student has 50 points. 'Free Coffee' reward costs 30 points                                | 1. Navigate to rewards section<br>2. Select 'Free Coffee' reward<br>3. Click redeem<br>4. Confirm redemption   | Points deducted                                          | Success message appear 'Reward redeemed'. Points balance reduced to 20         |                                                                           | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   |            | Cycle1: Check-in system not implemented<br><br>Cycle3: Postponed due to time limitation |
| TC_CRS_009 | Verify different locations give correct points                   | Functional                | Positive | Library check-in gives 10 points. Cafeteria check-in gives 5 points                        | 1. Check-in at Library<br>2. Check received points<br>3. Check-in at Cafeteria<br>4. Check received points     | -                                                        | Library gives 10 points and Cafeteria gives 5 points                           |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_010 | Verify points persist after logout and login                     | Security - Data           | Positive | Student has 200 points                                                                     | 1. Note points balance<br>2. Logout from application<br>3. Log back in<br>4. Verify points balance             | -                                                        | Points balance shows as 200                                                    |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_011 | Verify student cannot exceed daily check-in limit                | Security - Business logic | Positive | Student is logged in. Daily check-in limit 3. Student has already checked in 3 times today | 1. Attempt 4th check-in at valid location<br>2. Scan valid QR code<br>3. Observe system response               | No 4th check-in recorded. Points not awarded             | Error appears 'Daily check-in limit reached. Come back tomorrow'               |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_012 | Verify insufficient points prevents redemption                   | Functional                | Positive | Student has 5 points. Chosen reward costs 20 points                                        | 1. Navigate to reward section<br>2. Select chosen reward from reward list<br>3. Click redeem                   | No points deducted. Reward not awarded                   | Error appears 'Insufficient points balance'                                    |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_013 | Verify check-in counter resets after midnight                    | Functional                | Positive | Student checked 3 times on Monday. Time is now Tuesday 12.01 AM                            | 1. Attempt check-in on Tuesday<br>2. Scan valid QR code<br>3. Observe response                                 | Check-in successful                                      | Check-in successful                                                            |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |
| TC_CRS_014 | Verify check-in fails with invalid QR code                       | Security - Input          | Negative | Student is logged in                                                                       | 1. Open QR scanner<br>2. Scan invalid QR code<br>3. Observe response                                           | No check-in recorded                                     | Error appears 'Invalid QR code'                                                |                                                                           | Not Implemented         | Not Implemented            | Blocked                    | Blocked                    |            | Cycle1: Check-in system not implemented                                                 |

|  |            |                                                    |                           |          |                                                                      |                                                                                                  |                              |                                                                        |  |                 |                 |         |         |  |                                         |
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|  | TC_CRS_015 | Verify check-in fails at invalid location          | Security - Business logic | Negative | Student is logged in. Student scans QR code from off-campus location | 1. Scan QR code at invalid location<br>2. Attempt check-in                                       | No check-in recorded         | Check-in rejects. Error appears 'Not a valid check-in location'        |  | Not Implemented | Not Implemented | Blocked | Blocked |  | Cycle1: Check-in system not implemented |
|  | TC_CRS_016 | Verify redemption fails with insufficient points   | Functional                | Negative | Student has 15 points. Chosen reward costs 30 points                 | 1. Navigate to reward section<br>2. Select chosen reward from reward list<br>3. Click redeem     | Points balance remains as 15 | Error appears 'Insufficient points balance'                            |  | Not Implemented | Not Implemented | Blocked | Blocked |  | Cycle1: Check-in system not implemented |
|  | TC_CRS_017 | Verify redemption fails for unavailable reward     | Functional                | Negative | Student has 100 points. Chosen reward is currently unavailable       | 1. Navigate to reward section<br>2. Select chosen reward<br>3. Click if redeem button is enabled | No points deducted           | Redeem button is disabled or shows 'This reward currently unavailable' |  | Not Implemented | Not Implemented | Blocked | Blocked |  | Cycle1: Check-in system not implemented |
|  | TC_CRS_018 | Verify check-in fails when database is unavailable | Security - Config         | Negative | Student is logged in. Database connection is down                    | 1. Scan valid QR code<br>2. Attempt check-in<br>3. Observe response                              | Restore database connection  | Error appears 'System unavailable'. No points awarded                  |  | Not Implemented | Not Implemented | Blocked | Blocked |  | Cycle1: Check-in system not implemented |

| Test_ID    | Description                                                                                        | Test Category    | Type     | Pre-conditions                                                                        | Steps                                                                                                                                            | Post-conditions               | Expected results                                      | Actual results                                             | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID     | Notes                                                                                                                                                                     |
|------------|----------------------------------------------------------------------------------------------------|------------------|----------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------|------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TC_UPA-001 | Verify profile creation after registration                                                         | Functional       | Positive | Student is registered and logged in.                                                  | 1. Navigate to profile section<br>2. Check if profile is created<br>3. Verify details shown correctly                                            | -                             | Profile created successfully with registered details  | Profile created with details                               | Blocked                 | Fail                       | Fail                       | Fail                       | BUG_UPA_03 | Cycle1: Database integration for the profile not yet implemented                                                                                                          |
| TC_UPA-002 | Verify all data are accurate                                                                       | Functional       | Positive | Student profile contains name, email, bio                                             | 1. Navigate to profile section<br>2. View all displayed details<br>3. Verify them                                                                | -                             | All details are accurate                              | Cannot verify because some fields are missing              | Blocked                 | Fail                       | Fail                       | Fail                       | BUG_UPA_04 | Cycle1: Database integration for the profile not yet implemented                                                                                                          |
| TC_UPA-003 | Verify data cannot be manipulated                                                                  | Security - Data  | Positive | Student is logged in. Student A is viewing their own profile. Student B has a profile | 1. Attempt to manipulate URL to access user B's profile<br>2. Attempt to send API requests to change user B's data<br>3. Observe system response | No unauthorized changes made  | System blocks access. Returns 'Access Denied'         |                                                            | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1,2: Data manipulation cannot verify with static html url. Need a running environment                                                                                |
| TC_UPA-004 | Verify student can view their profiles                                                             | Functional       | Positive | Student is logged in. Student has a complete profile                                  | 1. Navigate to profile section<br>2. Observe displayed information                                                                               | -                             | Student sees their own profile with details           | Student can view their profile but partially displays data | Not Implemented         | Fail                       | Fail                       | Fail                       | BUG_UPA_03 | Cycle1: Profile exists as a static page and does not fetch dynamic data                                                                                                   |
| TC_UPA-005 | Verify student can update profile details                                                          | Functional       | Positive | Student is logged in                                                                  | 1. Navigate to profile section<br>2. Edit few details<br>3. Save changes                                                                         | Details updated in database   | Success message appear 'Profile saved successfully'   | Success message appears but details not update in database | Fail                    | Fail                       | Fail                       | Fail                       | BUG_UPA_01 | Cycle2: Update not works due to enrolled year null being sent and failed to load department id to choose<br>Cycle2: Update not works due to enrolled year null being sent |
| TC_UPA-006 | Verify data persists after logout and login                                                        | Functional       | Positive | Student updates name field                                                            | 1. Update name field<br>2. logout<br>3. log back in<br>4. View profile name                                                                      | -                             | Profile name still shows updated name correctly       |                                                            | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1: Database integration for the profile not yet implemented<br>Cycle2: Update not works due to enrolled year null being sent                                         |
| TC_UPA-007 | Verify profile picture can be uploaded                                                             | Functional       | Positive | Student is logged in                                                                  | 1. Navigate to profile section<br>2. Click upload picture<br>3. Select valid image<br>4. Save changes                                            | Profile picture saved         | Profile picture uploded, saved and displays correctly | Change photo button lets pick an image but doesn't save it | Fail                    | Fail                       | Fail                       | Pass                       | BUG_UPA_02 | Cycle2: Frontend failed to send backend required part.                                                                                                                    |
| TC_UPA-008 | Verify unauthorized user cannot view another student's profile                                     | Security - Auth  | Negative | Student A is logged in. Student B has a profile                                       | 1. User A attempts to access user B's profile via URL manipulation<br>2. Observe system response                                                 | User B's data not exposed     | Error message appears 'Access Denied'                 |                                                            | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1,2: Cannot test unauthorized access with static html url. Need a running environment                                                                                |
| TC_UPA-009 | Verify unauthorized user cannot edit another student's profile (Broken Object Level Authorization) | Security - Auth  | Negative | Student A is logged in. Student A knows student B's student id                        | 1. Student A attempts to send API requests to update student B's name<br>2. Observe system response                                              | Student B's profile unchanged | API returns 403 error or 'Unauthorized'               | Returns 403 Forbidden error                                | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                                                                         |
| TC_UPA-010 | Verify system rejects SQL injection in profile update fields (Insecure Direct Object Reference)    | Security - Input | Negative | Student is logged in                                                                  | 1. Navigate to profile section<br>2. Enter ' OR '1'='1 in name field<br>3. Attempt to save                                                       | No SQL injection executed     | System prevents SQL injection                         |                                                            | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1: Database integration for the profile not yet implemented<br>Cycle2: Profile update not work                                                                       |

| Test_ID    | Description                                                                                        | Test Category     | Type     | Pre-conditions                                                                  | Steps                                                                                                                                                                                                                                  | Post-conditions             | Expected results                                                                                  | Actual results                                                                                      | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID     | Notes                                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------|-------------------|----------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| TC_DMS_001 | Verify password stored securely (Weak storage cryptography)                                        | Security - Data   | Positive | Student profile created with password 'MyPass123'.                              | 1. Access database directly<br>2. Check password field                                                                                                                                                                                 | -                           | Password is hashed no plain text. Other sensitive data properly encrypted                         | Password is encrypted and securely stores                                                           | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                             |
| TC_DMS_002 | Verify student ID is masked in all displayed content (Personally identifiable information leakage) | Security - Data   | Positive | Student is logged in. Student ID is 'S123456'                                   | 1. Navigate to profile page, rewards page, check-in history page or any other page<br>2. Check how student ID is displayed                                                                                                             | -                           | Student ID is masked on all displayed content                                                     | Student ID is not displayed anywhere                                                                | Blocked                 | Pass                       | Pass                       | Fail                       | BUG_DMS_04 | Cycle1: Profile section is not fully built. Student Id is not displayed anywhere else to verify masking                       |
| TC_DMS_003 | Verify sensitive data is not exposed (Sensitive data leakage)                                      | Security - Data   | Positive | Student is logged in. Browser developer tools are open                          | 1. Navigate through application<br>2. Check browser console for any errors exposing data<br>3. Check network tab for API response<br>4. Check HTML source code for hidden sensitive data<br>5. Check local storage and session storage | -                           | No sensitive data appears in console, network, source or storage                                  | Student's username, password, JWT token, and role exposed across payload and preview in network tab | Fail                    | Fail                       | Fail                       | Fail                       | BUG_DMS_01 | -                                                                                                                             |
| TC_DMS_004 | Verify system is protected from unauthorized users (Broken Access Control)                         | Security - Auth   | Positive | Student is logged in. Student B is not logged in. Student A knows URL endpoints | 1. Student A is logs out<br>2. Try to access protected pages without login<br>3. Try to access API endpoints directly (eg. /api/profile)<br>4. Observe system response                                                                 | -                           | All protected pages redirect to login. API endpoints return '401 Unauthorized' or '403 Forbidden' |                                                                                                     | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1.2: Cannot test unauthorized access with static html url. Need a running environment                                    |
| TC_DMS_005 | Verify points balance cannot be manipulated (Integrity violation)                                  | Security - Data   | Positive | Student is logged in. Student has 120 points. Browser developer tools are open  | 1. Attempt to modify points balance via browser console<br>2. Attempt to modify API requests to change points balance<br>3. Attempt to send direct API request with manipulated<br>4. Check points balance after each attempt          | -                           | All manipulation attempts fails. Points balance remains 120                                       |                                                                                                     | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -          | Cycle1: No logic defined for points system and no database integration<br><br>Cycle2: Points balance is not displayed to test |
| TC_DMS_006 | Verify session token is secure (Session hijacking risk)                                            | Security - Auth   | Positive | Student is logged in                                                            | 1. Login successfully<br>2. Open browser developer tools<br>3. Go to application or storage tab<br>4. Check cookies for session token<br>5. Inspect cookie attributes                                                                  | -                           | Session token stored in browser cookies with HttpOnly, Secure, and Samesite attributes enabled    |                                                                                                     | Not Implemented         | Not Implemented            | Deferred                   | Deferred                   | -          | Postponed due to time limitation                                                                                              |
| TC_DMS_007 | Verify API endpoints require authentication                                                        | Security - Auth   | Positive | No user is logged in. List of API endpoints available                           | 1. Use API testing tool<br>2. Send requests to each endpoints without authentication token<br>3.Observe response codes                                                                                                                 | -                           | All endpoints return 401 Unauthorized or 403 Forbidden                                            | All endpoints return 403 Forbidden                                                                  | Pass                    | Pass                       | Pass                       | Pass                       | -          | -                                                                                                                             |
| TC_DMS_008 | Verify input sanitization on all user inputs (Injection flaw)                                      | Security - Input  | Positive | Student is opened the application                                               | 1. Enter special characters in all input fields<br>2. Enter HTML tags (eg: <b>test</b>)<br>3. Enter JavaScript alert('xss')<br>4. Submit and observe                                                                                   | -                           | All inputs are sanitized. Data displays as plain text                                             | The input displays as a plain text without executing any code                                       | Pass                    | Pass                       | Pass                       | Pass                       | BUG_DMS_02 | -                                                                                                                             |
| TC_DMS_009 | Verify unauthorized user cannot access sensitive data (Insecure Direct Object Reference)           | Security - Auth   | Negative | Student A is logged in. Student A has valid session token                       | 1. Student A attempts to access student B's profile via API<br>2. Use Student A's token to request (/api/profile?user_id=studentB_id)<br>3. Observe response                                                                           | -                           | API returns 403 Forbidden                                                                         | The endpoint uses /me to extract tudent's identity directly from authenticated session token.       | Pass                    | Pass                       | Pass                       | Pass                       | -          |                                                                                                                               |
| TC_DMS_010 | Verify sensitive data not exposed in error message                                                 | Security - Config | Negative | Student is not logged in                                                        | 1. Trigger errors like invalid login, invalid check-in<br>2. Observe error messages displayed<br>3. Check if any sensitive data is exposed (Student ID, Password)                                                                      | -                           | All error messages are generic. No sensitive data exposed                                         | No error messages displayed at all for invalid attempts                                             | Fail                    | Pass                       | Pass                       | Pass                       | BUG_DMS_03 | Cycle2: Error messages are generic and do not expose sensitive data                                                           |
| TC_DMS_011 | Verify database injection prevented (SQL Injection)                                                | Security - Input  | Negative | Student is logged in                                                            | 1. Enter SQL injection in all input fields ' OR '1'='1'<br>2. Enter NoSQL injection {'\$ne': null}<br>3. Submit and observe                                                                                                            | No unauthorized data access | All injection attempts rejected. No data leak                                                     | SQL injection attempts rejected                                                                     | Pass                    | Pass                       | Pass                       | Pass                       | -          | Cycle1: SQL injection attempts rejected in login and signup. Other input field not yet testable                               |

| Test ID   | Description                                                               | Test Category | Type     | Pre-conditions                                                               | Steps                                                                                                                                                                | Post-conditions               | Expected results                                                                   | Actual results                                                    | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID | Notes                                                                                                                                                                        |
|-----------|---------------------------------------------------------------------------|---------------|----------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TC_RA_001 | Verify system handles database failure without crash                      | Reliability   | Positive | Application is running. Student is logged in.                                | 1. Navigate to a section that reads from database (eg: profile, check-in history)<br>2. Disconnect or stop database<br>3. Observe application behavior               | Restore database connection   | Application displays cached data or graceful errors without crashing               | Application displays each previous data without crashing          | Not Implemented         | Blocked                    | Pass                       | Pass                       | -      | Cycle1: Application currently uses static data for most features. Cannot verify database failure handling<br><br>Cycle2: Cannot test, currently data not fetch from database |
| TC_RA_002 | Verify recently cached data is shown                                      | Reliability   | Positive | Internet connection is disabled                                              | 1. Disable internet connection<br>2. Open all pages<br>3. Observe displayed data<br>4. Check if data matches previously loaded data                                  | Re-enable internet connection | Most recently cached data is displayed                                             | Previously fetched data displays                                  | Not Implemented         | Not Implemented            | Pass                       | Pass                       | -      | Cycle1: No caching mechanism implemented.                                                                                                                                    |
| TC_RA_003 | Verify system recovers automatically when database connection established | Reliability   | Positive | Database is down. System currently shows cached data or error message        | 1. Observe current state<br>2. Restore database connection<br>3. Refresh the application<br>4. Observe data update                                                   | Database connection restored  | System automatically fetched new data from database                                | System automatically fetched new data from database               | Not Implemented         | Blocked                    | Pass                       | Pass                       | -      | Cycle1: Requires database failure and recovery logic<br><br>Cycle2: Cannot test, currently data not fetch from database                                                      |
| TC_RA_004 | Verify system handles partial database data                               | Reliability   | Positive | Database returns event data with missing fields like event name and location | 1. Navigate to events & updates section<br>2. Observe displayed events data<br>3. Check for any crashes or blank sections                                            | -                             | Available data is displayed. Missing fields handle gracefully without crashing     | Database added Not Null constraints to handle empty data properly | Not Implemented         | Blocked                    | Pass                       | Pass                       | -      | Cycle1: Database integration for dynamic content is incomplete<br><br>Cycle2: Cannot test, currently data not fetch from database                                            |
| TC_RA_005 | Verify system handles complete database outage                            | Reliability   | Negative | Applications is running. Database connection is disconnected                 | 1. Simulate database outage<br>2. Attempt to log in<br>3. Attempt to check-in<br>4. Attempt to view profile<br>5. Observe system behavior                            | Restore database connection   | Systems does not crash or expose database errors                                   | Systems does not crash previously fetched data shows              | Blocked                 | Blocked                    | Pass                       | Pass                       | -      | Cycle1.2: Database outage simulation cannot tested. Check-in and profile page not implemented                                                                                |
| TC_RA_006 | Verify system handles 500 server error smoothly                           | Reliability   | Negative | Server endpoint is configured to return HTTP 500 error                       | 1. Navigate to section that fails endpoints<br>2. Observe system response<br>3. Check if cached data is shown                                                        | Remove 500 error simulation   | System shows error message "Server error. Please try again later" without crashing |                                                                   | Not Applicable          | Not Applicable             | Not Applicable             | Not Applicable             | -      | Cycle1.2: No server errors observed during testing.                                                                                                                          |
| TC_RA_007 | Verify system handles sudden network disconnection during operation       | Reliability   | Negative | Student is actively using the app                                            | 1. Perform check-in operation<br>2. During database request, disconnect network<br>3. Observe system response<br>4. Reconnect network<br>5. Verify no corrupted data | Reconnect network             | System shows error message. No corrupted data saved                                |                                                                   | Blocked                 | Blocked                    | Blocked                    | Blocked                    | -      | Cycle1.2: Check-in feature not implemented yet                                                                                                                               |

|  | Module              | Test_ID       | Description                                | Test Category | Inputs                                                                                            | Type     | Pre-conditions              | Steps                                                                                                                                           | Post-conditions                        | Expected results                                                                                              | Actual results                                                                                                         | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Cycle4 Status (Regression) | Bug ID         | Notes                                        |
|--|---------------------|---------------|--------------------------------------------|---------------|---------------------------------------------------------------------------------------------------|----------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|----------------------------|----------------------------|----------------|----------------------------------------------|
|  | Club Hub            | TC_CH_STF_001 | Successfully create a new club             | Functional    | Club Name: Music Club<br>Status: true                                                             | Positive | Admin logged in             | 1. Enter club name<br>2. Enter status<br>3. Click save                                                                                          | Club visible in database and table     | "Club saved successfully" appears. New club appears in "Active Campus Societies Directory" table              | Failed to save data                                                                                                    | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CH_STF_01  | Cycle2: Failed to fetch data                 |
|  |                     | TC_CH_STF_002 | Update existing club                       | Functional    | Club Name: Music Club<br>Status: false                                                            | Positive | "Music Club" already exist. | 1. Change club status<br>2. Click save                                                                                                          | Club updated in database and table     | Club update correctly and appears updated details in table                                                    |                                                                                                                        | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -              | Cycle2: Failed to fetch data                 |
|  |                     | TC_CH_STF_003 | Delete existing club                       | Functional    | -                                                                                                 | Positive | "Music Club" already exist. | 1. Click remove item in chosen club record                                                                                                      | Club removed from table and database   | Club removed successfully                                                                                     | Failed to remove due to foreign key constraints                                                                        | Not Implemented         | Blocked                    | Fail                       | Pass                       | BUG_CH_STF_02  | Cycle2: Failed to fetch data                 |
|  |                     | TC_CH_STF_004 | Create a club with duplicate name          | Functional    | Club Name: Music Club<br>Status: false                                                            | Negative | -                           | 1. Enter already exist club name<br>2. Enter club status<br>3. Click save                                                                       | No duplicate club record created       | Prevent duplicate entries. Displays "Club name already exist"                                                 |                                                                                                                        | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -              | Cycle2: Failed to fetch data                 |
|  |                     | TC_CH_STF_005 | Create club with empty name                | Functional    | Club Name: empty<br>Status: false                                                                 | Negative | -                           | 1. Leave club name<br>2. Enter valid status                                                                                                     | -                                      | Displays "Club name required" or similar                                                                      |                                                                                                                        | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -              | Cycle2: Failed to fetch data                 |
|  |                     | TC_CH_STF_006 | Create club with invalid status            | Functional    | Club Name: Dance Club<br>Status: maybe                                                            | Negative | -                           | 1. Save club with invalid status                                                                                                                | -                                      | Displays validation error                                                                                     |                                                                                                                        | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -              | Cycle2: Failed to fetch data                 |
|  |                     |               |                                            |               |                                                                                                   |          |                             |                                                                                                                                                 |                                        |                                                                                                               |                                                                                                                        |                         |                            |                            |                            |                |                                              |
|  | Rewards Matrix      | TC_RS_STF_001 | Successfully create a new reward           | Functional    | Reward ID: DISC_5<br>Reward Name: 10% discount<br>Points: 1000<br>Discount: 10                    | Positive | Admin logged in             | 1. Enter Reward ID<br>2. Enter Reward Name<br>3. Enter Points Required<br>4. Enter Discount Value %<br>5. Click save                            | Reward visible in database and table   | "Reward saved successfully" appears. New reward appears in "Available Loyalty Rewards Items" table            | "Reward listing published" displayed. New reward appeared after few seconds in "Available Loyalty Rewards Items" table | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | Cycle2: Failed to fetch check-in and rewards |
|  |                     | TC_RS_STF_002 | Update existing reward                     | Functional    | Reward ID: DISC_6<br>Reward Name: 10% discount<br>Points: 200<br>Discount: 10                     | Positive | "DISC_5" reward exists      | 1. Click update on existing reward<br>2. Change Points from 1000 to 200<br>3. Click Save                                                        | -                                      | Points update correctly and appears updated details in table                                                  | Update button not working                                                                                              | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CRS_STF_01 | Cycle2: Failed to fetch check-in and rewards |
|  |                     | TC_RS_STF_003 | Delete existing reward                     | Functional    | -                                                                                                 | Positive | "DISC_5" reward exists      | 1. Click remove icon in chosen reward record                                                                                                    | Reward removed from table and database | Reward removed successfully                                                                                   | Reward removed successfully from table and database                                                                    | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | Cycle2: Failed to fetch check-in and rewards |
|  |                     | TC_RS_STF_004 | Save a reward with duplicate Id            | Functional    | Reward ID: DISC_1                                                                                 | Negative | -                           | 1. Enter already exist reward ID<br>2. Enter other fields properly<br>3. Click save                                                             | No duplicate reward record created     | Prevent duplicate entries. Displays "Reward ID already exist"                                                 | System silently updated existing reward with new details                                                               | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_CRS_STF_02 | Cycle2: Failed to fetch check-in and rewards |
|  |                     | TC_RS_STF_005 | Save reward with empty reward name         | Functional    | Reward ID: DISC_6<br>Reward Name: Points: 200<br>Discount: 10                                     | Negative | -                           | 1. Leave reward name<br>2. Click save                                                                                                           | -                                      | Displays "Reward name required" or similar                                                                    | Creating failed correctly and display alert                                                                            | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | Cycle2: Failed to fetch check-in and rewards |
|  |                     |               |                                            |               |                                                                                                   |          |                             |                                                                                                                                                 |                                        |                                                                                                               |                                                                                                                        |                         |                            |                            |                            |                |                                              |
|  | Facilities & Events | TC_FE_STF_001 | Successfully create a new department       | Functional    | Dept ID: D005<br>Dept Name: Engineering                                                           | Positive | Admin logged in             | 1. Enter Department ID<br>2. Enter Department Title Name<br>3. Click save                                                                       | Department visible in table            | Displays "Department saved successfully" and new department appears in "Registered Campus Tracks Block" table | Displayed "Department successfully saved" and appear in table after few seconds                                        | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | -                                            |
|  |                     | TC_FE_STF_002 | Update existing department                 | Functional    | Dept ID: D005<br>Dept Name: Aviation                                                              | Positive | Department "D005" exists    | 1. Select department record and click update<br>2. Enter new department name<br>3. Click save                                                   | Department updated in table            | Displays "Department updated successfully" and updated data appears in "Registered Campus Tracks Block" table | Department name updated correctly in both table and database                                                           | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | -                                            |
|  |                     | TC_FE_STF_003 | Successfully create a new event            | Functional    | Event: Handawa<br>Date: 2026-12-15<br>Time: 18:30<br>Venue: Car Park<br>Committee: Rotract Club   | Positive | Department "D005" exists    | 1. Enter Event Name<br>2. Select Scheduled Date<br>3. Select Scheduled Time<br>4. Enter Venue<br>5. Enter Organizing Committee<br>6. Click save | Event visible in table                 | Success message appears. New event appears in "Staged Campus Events Calendar" table with unique ID            | Failed to create a new event                                                                                           | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_FS_STF_01  | wrong event title                            |
|  |                     | TC_FE_STF_004 | Update existing event                      | Functional    | Event: Handawa<br>Date: 2026-12-15<br>Time: 17:30<br>Venue: Car Park<br>Committee: Rotract Club   | Positive | Event "Handawa" exists      | 1. Enter new event time<br>2. Keep other data<br>3. Click save                                                                                  | Event updated in table                 | Displays "Event updated successfully" and updated data appears in "Staged Campus Events Calendar" table       | Failed to update due to a database constraints failure                                                                 | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_FS_STF_02  | -                                            |
|  |                     | TC_FE_STF_005 | Delete existing event                      | Functional    | -                                                                                                 | Positive | -                           | 1. Click remove icon in chosen event record                                                                                                     | -                                      | Delete successfully from the table and database                                                               | Successfully deleted from table and database                                                                           | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | -                                            |
|  |                     | TC_FE_STF_006 | Create a new department with empty details | Functional    | Dept ID: empty<br>Dept Name: empty                                                                | Negative | -                           | 1. Click save with empty fields                                                                                                                 | No new record created                  | Validation error like "Department ID and Name are required" appears                                           | Displays "Please fill out both Department ID and Department Name!"                                                     | Not Implemented         | Blocked                    | Pass                       | Pass                       | -              | -                                            |
|  |                     | TC_FE_STF_007 | Create a new department with duplicate ID  | Functional    | Dept ID: D005<br>Dept Name: Tourism                                                               | Negative | Department "D005" exists    | 1. Enter valid details in other fields<br>2. Enter duplicate departmetn id in id field<br>3. Click save                                         | No new record created                  | Displays "Department ID already exist" or similar                                                             | Updated silently existing department without displaying error message                                                  | Not Implemented         | Blocked                    | Fail                       | Fail                       | BUG_FS_STF_03  | -                                            |
|  |                     | TC_FE_STF_008 | Create a new event with past date          | Functional    | Event: Initium 3.0<br>Date: 2020-11-05<br>Time: 11:30<br>Venue: Auditorium<br>Committee: Leo Club | Negative | -                           | 1. Enter valid details in other fields<br>2. Enter past date<br>3. Click save                                                                   | -                                      | Event not created with past date                                                                              |                                                                                                                        | Not Implemented         | Blocked                    | Blocked                    | Blocked                    | -              | -                                            |

|  |  |               |                                          |            |                                                                                                                                                            |          |                                      |                                                                                                                                                                                                   |                                               |                                                                                                  |                                                                                                    |                 |         |         |         |               |   |
|--|--|---------------|------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------|---------|---------|---------|---------------|---|
|  |  | TC_FE_STF_009 | Save event with empty required fields    | Functional | Event: empty<br>Date: 2020-11-05<br>Time: 11:30<br>Venue: Auditorium<br>Committee: Leo Club                                                                | Negative | -                                    | 1. Leave event name as empty<br>2. Fill other fields<br>3. Click save                                                                                                                             | -                                             | Failed to save event with missing required name                                                  |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_FE_STF_010 | Save event with invalid date format      | Functional | Event: Initium 3.0<br>Date: 12/09/2026<br>Time: 11:30<br>Venue: Auditorium<br>Committee: Leo Club                                                          | Negative | -                                    | 1. Enter valid details in other fields<br>2. Enter date with invalid format<br>3. Click save                                                                                                      | -                                             | Event not saved with invalid date format                                                         |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_FE_STF_011 | Cancel deletion confirmation             | Functional | Event: Handawa<br>Date: 2026-12-15<br>Time: 11:30<br>Venue: Car Park<br>Committee: Roctract Club                                                           | Negative | -                                    | 1. Click remove icon in event record<br>2. Cancel confirmation dialog                                                                                                                             | -                                             | Data remains in table                                                                            | Data not deleted and remains                                                                       | Not Implemented | Blocked | Pass    | Pass    | -             | - |
|  |  |               |                                          |            |                                                                                                                                                            |          |                                      |                                                                                                                                                                                                   |                                               |                                                                                                  |                                                                                                    |                 |         |         |         |               |   |
|  |  | TC_LC_STF_001 | Successfully create a new location       | Functional | Location ID: AUDI_01<br>Name: Main Auditorium<br>Capacity: 1000                                                                                            | Positive | Admin logged in                      | 1. Enter Location ID<br>2. Enter Location Name<br>3. Enter Capacity<br>4. Click [Save Location]                                                                                                   | Location visible with Live Operational Status | Success message appears and location appears in "Tracked Infrastructure Location Matrices" table | Success message appeared and location appeared in "Tracked Infrastructure Location Matrices" table | Not Implemented | Blocked | Pass    | Pass    | -             | - |
|  |  | TC_LC_STF_002 | Update Location Status                   | Functional | -                                                                                                                                                          | Positive | Location exists                      | 1. Click Actions (Edit)<br>2. Change Live Operational Status<br>3. Save                                                                                                                           | -                                             | Status updated successfully                                                                      | Edit action not working                                                                            | Not Implemented | Blocked | Fail    | Fail    | BUG_LC_STF_01 | - |
|  |  | TC_LC_STF_003 | Delete Location                          | Functional | -                                                                                                                                                          | Positive | Location exists with no dependencies | 1. Click Delete<br>2. Confirm                                                                                                                                                                     | -                                             | Location removed from table                                                                      | Removed successfully                                                                               | Not Implemented | Blocked | Pass    | Pass    | -             | - |
|  |  | TC_LC_STF_004 | Save location with duplicate ID          | Functional | Location ID: GYM_01<br>Name: Fitness<br>Capacity: 100                                                                                                      | Negative | -                                    | 1. Save location with duplicate Location ID                                                                                                                                                       | -                                             | Displays "Location ID already exists"                                                            | Updated silently existing location without displaying error message                                | Not Implemented | Blocked | Fail    | Fail    | BUG_LC_STF_02 | - |
|  |  | TC_LC_STF_005 | Save capacity as a text                  | Functional | Capacity: infinite                                                                                                                                         | Negative | -                                    | 1. Save location with Capacity as text                                                                                                                                                            | -                                             | Validation error displays as "Capacity must be a number" or similar                              | Location not created. Failed to display error message                                              | Not Implemented | Blocked | Fail    | Fail    | BUG_LC_STF_03 | - |
|  |  | TC_LC_STF_006 | Save location with empty name            | Functional | -                                                                                                                                                          | Negative | -                                    | 1. Save location with empty Location Name                                                                                                                                                         | -                                             | Displays "Location Name is required"                                                             | Displays "Please fill out all fields!"                                                             | Not Implemented | Blocked | Pass    | Pass    | -             | - |
|  |  |               |                                          |            |                                                                                                                                                            |          |                                      |                                                                                                                                                                                                   |                                               |                                                                                                  |                                                                                                    |                 |         |         |         |               |   |
|  |  | TC_LC_TM_001  | Successfully create new bus              | Functional | Bus ID: BUS_003<br>Bus Number: WP NB-5241<br>Status: ON_TIME<br>Departure: Gampaha<br>Arrival: Main Campus<br>Departure Time: 07:30<br>Arrival Time: 08:45 | Positive | Admin logged in                      | 1. Enter Bus ID<br>2. Enter Bus Number<br>3. Select Current Status<br>4. Select Departure Location<br>5. Select Arrival Location<br>6. Set Departure Time<br>7. Set Arrival Time<br>8. Click save | Bus visible with all details                  | Success message appears and bus appears in "Active Transit Telemetry Grid" table                 | Failed to save data                                                                                | Not Implemented | Blocked | Fail    | Fail    | BUG_TM_STF_01 | - |
|  |  | TC_LC_TM_002  | Update bus status                        | Functional | Bus ID: BUS_003<br>Bus Number: WP NB-5241<br>Status: DELAYED<br>Departure: Gampaha<br>Arrival: Main Campus<br>Departure Time: 07:30<br>Arrival Time: 08:45 | Positive | Bus exists                           | 1. Click Edit on existing bus<br>2. Change Status from ON_TIME to DELAYED<br>3. Save                                                                                                              | -                                             | Status updated in transit grid                                                                   |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_LC_TM_003  | Delete existing bus                      | Functional | -                                                                                                                                                          | Positive | Bus exists                           | 1. Click Delete<br>2. Confirm                                                                                                                                                                     | -                                             | Bus removed from transit grid                                                                    | Failed to remove                                                                                   | Not Implemented | Blocked | Fail    | Fail    | BUG_TM_STF_02 | - |
|  |  | TC_LC_TM_004  | Save bus with duplicate bus ID           | Functional | Bus ID: BUS_001                                                                                                                                            | Negative | Bus "BUS_001" already exists         | 1. Save bus with duplicate Bus ID                                                                                                                                                                 | -                                             | Error: "Bus ID already exists"                                                                   |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_LC_TM_005  | Save bus with empty bus number           | Functional | -                                                                                                                                                          | Negative | -                                    | 1. Save bus with empty Bus Number                                                                                                                                                                 | -                                             | Error: "Bus Number is required"                                                                  |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_LC_TM_006  | Save bus with invalid Status value       | Functional | Status: FLYING (not in dropdown options)                                                                                                                   | Negative | -                                    | 1. Save bus with invalid Status value                                                                                                                                                             | -                                             | Validation error or default to ON_TIME                                                           |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_LC_TM_007  | Save bus with missing Departure Location | Functional | -                                                                                                                                                          | Negative | -                                    | 1. Save bus with missing Departure Location                                                                                                                                                       | -                                             |                                                                                                  |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |
|  |  | TC_LC_TM_008  | Save departure time with invalid format  | Functional | Departure Time: 25:00                                                                                                                                      | Negative | Bus exists                           | 1. Edit bus<br>2. Set Departure Time to invalid format                                                                                                                                            | -                                             | Error: "Invalid time format. Use HH:MM"                                                          |                                                                                                    | Not Implemented | Blocked | Blocked | Blocked | -             | - |

|  | Test ID       | Testing Type | Dashboard | Test Scenario      | Test Steps                                    | Expected Result                         | Actual Result                                                                       | Status |
|--|---------------|--------------|-----------|--------------------|-----------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------|--------|
|  | TC_CHATBOT_01 | Smoke        | Both      | Chatbot loads      | 1. Login<br>2. Open dashboard student/staff   | Chatbot visible                         | Chatbot visible                                                                     | Pass   |
|  | TC_CHATBOT_02 | Smoke        | Both      | Send & Receive     | 1. Type "Hello"<br>2. Click send              | Receives a reply                        | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_03 | Smoke        | Both      | Factual Result     | 1. Ask how many kilometers from earth to moon | Reply contains 384,400 or 384,400km     | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_04 | Smoke        | Both      | API connectivity   | 1. Type any message                           | HTTP 200 response within 3 seconds      | HTTP 200 response receive within 3 seconds                                          | Pass   |
|  | TC_CHATBOT_05 | Functional   | Both      | Empty message      | 1. Click send with blank input                | No API call OR "Please enter a message" | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_06 | Functional   | Both      | Very long message  | 1. Type 1000+ characters                      | Response received (no timeout/crash)    | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_07 | Functional   | Both      | Special characters | 1. Type "@*\$^\$^#&()%"                       | No crash, returns some response         | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_08 | Functional   | Both      | Non-English        | 1. Type "Bonjour" or "नमस्ते"                 | Responds appropriately (if supported)   | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |
|  | TC_CHATBOT_09 | Functional   | Both      | Numbers only       | 1. Type "647453"                              | Responds (not just echo the number)     | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail   |

|  |               |            |      |           |                          |                                                   |                                                                                     |      |
|--|---------------|------------|------|-----------|--------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|------|
|  | TC_CHATBOT_10 | Functional | Both | Follow-up | 1. Ask "2+2?" then "3+3" | If stateful, understands about refers to addition | Displays Sorry, I'm having trouble connecting to the core server gateway right now. | Fail |
|--|---------------|------------|------|-----------|--------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|------|

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|  |                                | BUG_LOG_05 | Password exposed in network payload tab                                      | TC_LOG_STU_022 | Critical | High   | Frontend Developer                     | Open   | 1. Login with valid student details<br>2. Click login network request in dev tools<br>3. Observe password exposed in payload tab                                        | 15/05/2026 |            |
|  |                                |            |                                                                              | TC_LOG_STF_021 | Critical | High   | Frontend Developer                     | Open   | 1. Login with valid staff details<br>2. Click login network request in dev tools<br>3. Observe password exposed in payload tab                                          | 15/05/2026 |            |
|  |                                | BUG_LOG_06 | JWT token, username, role exposed in browser's dev tools                     | TC_LOG_STU_023 | Critical | High   | Frontend Developer                     | Open   | 1. Login with valid student details<br>2. Click login network request in dev tools<br>3. Observe token, username and role exposed in preview tab or localstorage        | 15/05/2026 |            |
|  |                                |            |                                                                              | TC_LOG_STF_022 | Critical | High   | Frontend Developer                     | Open   | 1. Login with valid staff details<br>2. Click login network request in dev tools<br>3. Observe token, username and role exposed in preview tab or localstorage          | 15/05/2026 |            |
|  |                                | BUG_LOG_07 | Staff redirects to same student dashboard. No separate admin dashboard exist | TC_LOG_STF_008 | Critical | High   | Frontend Developer & Backend Developer | Closed | 1. Login with valid staff Id and password<br>2. Observe successful login redirects to student dashboard                                                                 | 15/05/2026 | 25/05/2026 |
|  | Registration                   | BUG_REG_01 | New student's points balance not starts with zero                            | TC_REG_002     | Critical | High   | Backend Developer & Database Manager   | Open   | 1. Register new student with valid details<br>2. Observe points balance not starts with zero                                                                            | 15/05/2026 |            |
|  |                                | BUG_REG_02 | No validation for invalid email format                                       | TC_REG_008     | High     | High   | Frontend Developer & Backend Developer | Open   | 1. Register new student with invalid email format<br>2. Observe registration succeeds with no error and validation                                                      | 15/05/2026 |            |
|  |                                | BUG_REG_03 | No validation for weak passwords                                             | TC_REG_009     | High     | High   | Backend Developer & Frontend Developer | Open   | 1. Register new student with weak password<br>2. Observe registration succeeds with no error and validation                                                             | 15/05/2026 |            |
|  |                                | BUG_REG_04 | No error message for registration with duplicat student Id                   | TC_REG_011     | Medium   | High   | Backend Developer                      | Closed | 1. Try to register new student with already exist Id<br>2. Observe registration failed but no error displays                                                            | 15/05/2026 | 25/05/2026 |
|  |                                | BUG_REG_05 | No error message for registration with duplicat student email                | TC_REG_012     | Medium   | High   | Backend Developer                      | Closed | 1. Try to register new student with already exist email<br>2. Observe registration failed but no error displays                                                         | 15/05/2026 | 25/05/2026 |
|  |                                | BUG_REG_06 | All student's details exposed in network payload tab                         | TC_REG_016     | Critical | High   | Frontend Developer                     | Open   | 1. Signup with valid student details<br>2. Click signup network request in dev tools<br>3. Observe student details including Id, password, email exposed in preview tab | 16/05/2026 |            |
|  |                                | BUG_REG_07 | Student registration fails due to null ID being sent from frontend           | TC_REG_001     | Critical | High   | Frontend Developer                     | Open   | 1. Try to signup with valid student details<br>2. Click register button<br>3. Observe registration failed and console output in backend                                 | 22/05/2026 |            |
|  | User Profiles & Authentication | BUG_UPA_01 | Failed to store updated profile details in database                          | TC_UPA-005     | Medium   | Medium | Frontend Developer                     | Open   | 1. Update existing profile details<br>2. Click save changes<br>3. Observe feedback appears but details not stored                                                       | 16/05/2026 |            |
|  |                                | BUG_UPA_02 | Chosen image not saves as profile picture                                    | TC_UPA-007     | Low      | Low    | Frontend Developer                     | Closed | 1. Click change photo<br>2. Choose preferred image<br>3. Observe image not saved                                                                                        | 16/05/2026 | 25/05/2026 |

|  |                            |            |                                                                                            |            |          |        |                    |        |                                                                                                                                                                                                                                                                                                      |            |            |
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|  | Authentication             | BUG_UPA_03 | Profile displays only few fields. Failed to display all registered data except ID          | TC_UPA-001 | Medium   | Medium | Frontend Developer | Open   | 1. Navigate to profile section<br>2. Observe displayed data                                                                                                                                                                                                                                          | 22/05/2026 |            |
|  |                            | BUG_UPA_04 | Profile details partially displayed. Cannot verify accuracy of missing fields              | TC_UPA-002 | Medium   | Medium | Frontend Developer | Open   | 1. Navigate to profile section<br>2. Observe data in other fields missing                                                                                                                                                                                                                            | 22/05/2026 |            |
|  | Data Management & Security | BUG_DMS_01 | Sensitive data exposed across browser's network payload and preview tabs                   | TC_DMS_003 | Critical | High   | Frontend Developer | Open   | 1. Navigate to network tab<br>2. Click login or signup request<br>3. Check payload and preview tabs<br>3. Observe sensitive data is displayed                                                                                                                                                        | 17/05/2026 |            |
|  |                            | BUG_DMS_02 | Frontend does not sanitise/escape malicious inputs before sending to server                | TC_DMS_008 | Low      | Low    | Frontend Developer | Open   | 1. Open login page with developer tools<br>2. In network tab, check Preserve log<br>3. Enter valid id<br>4. Enter <img src=> as password<br>5. Click login button<br>6. Click login request in network tab<br>7. Check payload tab<br>8. Observe input not sanitized and not rejected before sending | 17/05/2026 |            |
|  |                            | BUG_DMS_03 | No error messages displays to notify users                                                 | TC_DMS_010 | High     | High   | Frontend Developer | Closed | 1. Perform invalid login<br>2. Observe no error message appears                                                                                                                                                                                                                                      | 17/05/2026 | 25/05/2026 |
|  |                            | BUG_DMS_04 | Student Id displays in profile section but not properly masked                             | TC_DMS_002 | Critical | High   | Frontend Developer | Open   | 1. Open profile section<br>2. Observe student ID displayed without masking                                                                                                                                                                                                                           | 25/05/2026 |            |
|  | Facility & Event Updates   | BUG_FEU_01 | Facility details display as raw JSON and undefined values instead of user-friendly details | TC_FEU_001 | High     | High   | Frontend Developer | Closed | 1. Navigate to facility status section<br>2. Observe data displayed as raw JSON format                                                                                                                                                                                                               | 22/05/2026 | 25/05/2026 |
|  |                            | BUG_FEU_02 | Events section failed to fetch and display available events with details                   | TC_FEU_004 | High     | Medium | Frontend Developer | Open   | 1. Navigate to events section<br>2. Observe data failed to fetch and display correctly                                                                                                                                                                                                               | 22/05/2026 |            |
|  |                            | BUG_FEU_03 | Facility statuses section refresh causes redirection to profile dashboard                  | TC_FEU_005 | High     | High   | Frontend Developer | Open   | 1. Navigate to facility status section<br>2. Observe page refresh automatically redirect to profile dashboard                                                                                                                                                                                        | 22/05/2026 |            |
|  |                            | BUG_FEU_04 | Past events displays with upcoming events. Failed to display seperately                    | TC_FEU_010 | Low      | Low    | Frontend Developer | Open   | 1. Navigate to facility status section<br>2. Check each event's date<br>3. Observe past events and upcoming events display without seperating                                                                                                                                                        | 23/05/2026 |            |
|  | Transit & Navigation       | BUG_TN_01  | System failed to fetch and display transit bus details from the database                   | TC_TN_001  | Critical | High   | Frontend Developer | Closed | 1. Navigate to transit and navigation section<br>2. Observe data failed to fetch and display correctly                                                                                                                                                                                               | 22/05/2026 | 25/05/2026 |
|  |                            | BUG_TN_02  | Transit & Navigation section refresh causes redirection to profile dashboard               | TC_TN_004  | High     | High   | Frontend Developer | Open   | 1. Navigate to transit and navigation section<br>2. Observe page refresh automatically redirect to profile dashboard                                                                                                                                                                                 | 23/05/2026 |            |
|  | Club Hub                   | BUG_CH_01  | System failed to fetch and display club details from the database                          | TC_CH_001  | Critical | High   | Frontend Developer | Closed | 1. Navigate to clubs section<br>2. Observe data failed to fetch and display correctly                                                                                                                                                                                                                | 22/05/2026 | 25/05/2026 |
|  |                            |            | Failed to display each club's status                                                       |            | High     | High   | Frontend Developer | Open   | 1. Navigate to clubs section<br>2. Observe active status                                                                                                                                                                                                                                             | 23/05/2026 |            |
|  |                            | BUG_CH_02  | Failed to display student joined club's name and joined date                               | TC_CH_002  | High     | High   | Frontend Developer | Open   | 1. Navigate to clubs section<br>2. Observe failed to display joined club's name, active status and joined date under My Enrolled Clubs section                                                                                                                                                       | 23/05/2026 |            |

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|  |                 |                | System displays other students's joined clubs as well                 |               | Critical | High | Frontend Developer | Open | 1. Navigate to clubs section<br>2. Check displayed data with database under My Enrolled Clubs section           | 23/05/2026 |  |
|  | Staff Dashboard | BUG_CH_STF_01  | Failed to save new club data                                          | TC_CH_STF_001 | High     | High | Frontend Developer | Open | 1. Navigate to Club Management<br>2. Enter new club details<br>3. Click Save<br>4. Observe failure to save data | 25/05/2026 |  |
|  |                 | BUG_CH_STF_02  | Student cannot leave a joined club due to wrong usage of HTTP method  | TC_CH_STF_003 | Medium   | High | Frontend Developer | Open | 1. Open joined clubs list<br>2. Attempt to leave a club<br>3. Request fails due to incorrect HTTP method usage  | 25/05/2026 |  |
|  |                 | BUG_CRS_STF_01 | Update button in reward matrix not working                            | TC_RS_STF_002 | High     | High | Frontend Developer | Open | 1. Open reward matrix<br>2. Click Update button                                                                 | 25/05/2026 |  |
|  |                 | BUG_CRS_STF_02 | System silently updated existing reward with new details              | TC_RS_STF_004 | High     | High | Frontend Developer | Open | 1. Update reward details<br>2. Save changes                                                                     | 25/05/2026 |  |
|  |                 | BUG_FS_STF_01  | Failed to create a new event                                          | TC_FE_STF_003 | High     | High | Frontend Developer | Open | 1. Navigate to Event creation<br>2. Fill event details<br>3. Click Create                                       | 25/05/2026 |  |
|  |                 | BUG_FS_STF_02  | Failed to update due to a database constraints failure                | TC_FE_STF_004 | High     | High | Frontend Developer | Open | 1. Update an event<br>2. Trigger database constraint condition (e.g., duplicate/invalid data)                   | 25/05/2026 |  |
|  |                 | BUG_FS_STF_03  | Updated silently existing department without displaying error message | TC_FE_STF_007 | Medium   | High | Frontend Developer | Open | 1. Edit department details in event module<br>2. Save changes                                                   | 25/05/2026 |  |
|  |                 | BUG_LC_STF_01  | Edit action in location control not working                           | TC_LC_STF_002 | High     | High | Frontend Developer | Open | 1. Navigate to Location Control<br>2. Try editing a location<br>3. Save changes                                 | 25/05/2026 |  |
|  |                 | BUG_LC_STF_02  | Updated silently existing location without displaying error message   | TC_LC_STF_004 | Medium   | High | Frontend Developer | Open | 1. Update location data<br>2. Save changes                                                                      | 25/05/2026 |  |
|  |                 | BUG_LC_STF_03  | Location not created. Failed to display error message                 | TC_LC_STF_005 | High     | High | Frontend Developer | Open | 1. Create a new location with invalid/duplicate data<br>2. Click save                                           | 25/05/2026 |  |
|  |                 | BUG_TM_STF_01  | Failed to save transit data                                           | TC_LC_TM_001  | High     | High | Frontend Developer | Open | 1. Go to Transit Management<br>2. Enter bus/transit details<br>3. Click Save                                    | 25/05/2026 |  |
|  |                 | BUG_TM_STF_02  | Failed to remove bus details                                          | TC_LC_TM_003  | High     | High | Frontend Developer | Open | 1. Open bus details list<br>2. Try to remove a bus entry<br>3. Deletion fails                                   | 25/05/2026 |  |

| Feature | Table Name  | Test ID       | HTTP Method | Endpoint                                           | Test Description         | Request Body                                                                                                                                                                                                                                                                                                                                                                                   | Expected Status | Cycle1 (Initial testing) | Cycle2 (Regression) | Cycle3 (Regression) | Bug ID         | Notes                                                                                     |
|---------|-------------|---------------|-------------|----------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------|---------------------|---------------------|----------------|-------------------------------------------------------------------------------------------|
|         | departments | TC_DBC_DEP_01 | GET         | /live-campus-hub/api/v1/departments                | Get all departments      | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_DEP_02 | GET         | /live-campus-hub/api/v1/departments/D001           | Get single department    | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_DEP_03 | POST        | /live-campus-hub/api/v1/departments                | Create new department    | {"department_id":"D001","department_Name":"Management"}                                                                                                                                                                                                                                                                                                                                        | 201 Created     | Pass                     | Pass                | Pass                | BUG_DBC_DEP_01 | Cycle2: Issue fixed                                                                       |
|         |             | TC_DBC_DEP_05 | PUT         | /live-campus-hub/api/v1/departments                | Update exist department  | {"department_id":"D001","department_Name":"Finance"}                                                                                                                                                                                                                                                                                                                                           | 204 No Content  | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_DEP_06 | DELETE      | /live-campus-hub/api/v1/departments                | Delete exist department  | {"department_id":"D002","department_Name":"Engineering"}                                                                                                                                                                                                                                                                                                                                       | 204 No Content  | Pass                     | Pass                | Pass                | -              | Cycle 2: Added {id} to the endpoint                                                       |
|         | students    | TC_DBC_STD_01 | GET         | /live-campus-hub/api/v1/students                   | Get all students         | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Fail                     | Pass                | Pass                | BUG_DBC_STD_01 | Cycle 2: Fixed 'enrolled_year' data type mismatch                                         |
|         |             | TC_DBC_STD_02 | GET         | /live-campus-hub/api/v1/students/me                | Get single student       | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Fail                     | Pass                | Pass                | BUG_DBC_STD_02 | Cycle2: Issue fixed                                                                       |
|         |             | TC_DBC_STD_03 | POST        | /live-campus-hub/api/v1/students                   | Create new student       | {"student_id":"S2406234","student_pwd":"2345671","first_name":"Nimasha","last_name":"Kalhan","phone":"715472527","email":"nimashak@gmail.com","enrolled_year":"2025","department_id":"D002"}                                                                                                                                                                                                   | 201 Created     | Fail                     | Pass                | Pass                | BUG_DBC_STD_03 | Cycle2: Issue fixed                                                                       |
|         |             | TC_DBC_STD_04 | POST        | /live-campus-hub/api/v1/students/save-students     | Create multiple students | [{"student_id":"STU_013","student_pwd":"student123#","first_name":"Heyoon","last_name":"Jeong","phone":"0716492748","email":"heyoonj@gmail.com","enrolled_year":"2023","department_id":"D001"}, {"student_id":"STU_014","student_pwd":"student123#","first_name":"Sina","last_name":"Deinert","phone":"0715472500","email":"deinert@gmail.com","enrolled_year":"2024","department_id":"D004"}] | 201 Created     | Blocked                  | Pass                | Pass                | -              | New in Cycle 2 (Not available in Cycle 1)                                                 |
|         |             | TC_DBC_STD_05 | PUT         | /live-campus-hub/api/v1/students/me                | Update exist student     | {"student_id": "STU_013", "student_pwd": "sanduni123#", "first_name": "Sanduni", "last_name": "Hansika", "phone": "0714224503", "email": "sanduni@gmail.com", "department_id": "D004"}                                                                                                                                                                                                         | 204 No Content  | Fail                     | Pass                | Pass                | BUG_DBC_STD_04 | Cycle 2: Issue fixed but password turned into a plain text after update student's details |
|         |             | TC_DBC_STD_06 | DELETE      | /live-campus-hub/api/v1/students/STU_014           | Delete exist student     | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 204 No Content  | Fail                     | Pass                | Pass                | BUG_DBC_STD_05 | Cycle2: Issue fixed                                                                       |
|         | rewards     | TC_DBC_RWD_01 | GET         | /live-campus-hub/api/v1/rewards                    | Get all rewards          | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_RWD_02 | GET         | /live-campus-hub/api/v1/rewards/DISC_15            | Get single reward        | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_RWD_03 | POST        | /live-campus-hub/api/v1/rewards                    | Create new reward        | {"reward_id":"DISC", "reward_name":"25% discount", "reward_points":"1080", "discount_percentage":"25"}                                                                                                                                                                                                                                                                                         | 201 Created     | Pass                     | Pass                | Pass                | BUG_DBC_RWD_01 | Cycle2: Issue fixed                                                                       |
|         |             | TC_DBC_RWD_04 | POST        | /live-campus-hub/api/v1/rewards/add-points(points) | Add points to student    | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 201 Created     | Blocked                  | Pass                | Pass                | -              | New in Cycle 2 (Not available in Cycle 1)                                                 |
|         |             | TC_DBC_RWD_05 | DELETE      | /live-campus-hub/api/v1/rewards/DISC_5             | Delete reward            | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 204 No Content  | Blocked                  | Pass                | Pass                | -              | New in Cycle 2 (Not available in Cycle 1)                                                 |
|         |             | TC_DBC_LOC_01 | GET         | /live-campus-hub/api/v1/locations                  | Get all locations        | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |
|         |             | TC_DBC_LOC_02 | GET         | /live-campus-hub/api/v1/locations/CAR_PARK         | Get single location      | N/A                                                                                                                                                                                                                                                                                                                                                                                            | 200 OK          | Pass                     | Pass                | Pass                | -              | -                                                                                         |

|                               |           |               |        |                                                                                      |                              |                                                                                                                                                  |                |         |      |      |                |                                                                                |
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| Database & Backend Connection | locations | TC_DBC_LOC_03 | GET    | /live-campus-hub/api/v1/locations/CAR_PARK/status                                    | Get single location's status | N/A                                                                                                                                              | 200 OK         | Blocked | Pass | Pass | -              | New in Cycle 2 (Not available in Cycle 1)                                      |
|                               |           | TC_DBC_LOC_04 | GET    | /live-campus-hub/api/v1/locations/all-status                                         | Get location status          | N/A                                                                                                                                              | 202 OK         | Blocked | Pass | Pass | -              | New in Cycle 2 (Not available in Cycle 1)                                      |
|                               |           | TC_DBC_LOC_05 | POST   | /live-campus-hub/api/v1/locations                                                    | Create new location          | {"location_id": "GYM_01", "location_name": "Gym", "capacity": "500"}                                                                             | 201 Created    | Pass    | Pass | Pass | BUG_DBC_LOC_01 | Cycle2: Issue fixed                                                            |
|                               |           | TC_DBC_LOC_06 | PUT    | /live-campus-hub/api/v1/locations                                                    | Update exist location        | {"location_id": "MAIN_CANTEEN", "location_name": "Main Cafeteria", "capacity": "1500"}                                                           | 204 No Content | Pass    | Pass | Pass | -              | -                                                                              |
|                               |           | TC_DBC_LOC_07 | DELETE | /live-campus-hub/api/v1/locations/OPEN_CANTEEN                                       | Delete exist location        | N/A                                                                                                                                              | 204 No Content | Pass    | Pass | Pass | -              | -                                                                              |
|                               | events    | TC_DBC_EVT_01 | GET    | /live-campus-hub/api/v1/events                                                       | Get all events               | N/A                                                                                                                                              | 200 OK         | Fail    | Pass | Pass | BUG_DBC_EVT_01 | Cycle2: Issue fixed                                                            |
|                               |           | TC_DBC_EVT_02 | GET    | /live-campus-hub/api/v1/events/search?title=Viramaya                                 | Get one event                | N/A                                                                                                                                              | 200 OK         | Blocked | Pass | Pass | -              | New in Cycle 2 (Not available in Cycle 1)                                      |
|                               |           | TC_DBC_EVT_03 | POST   | /live-campus-hub/api/v1/events                                                       | Create new event             | {"event_title": "Hackathon", "event_date": "2026-06-20", "event_time": "10:00:00", "location_name": "Main Cafeteria", "club_name": "Leo Club"}   | 201 Created    | Fail    | Fail | Pass | BUG_DBC_EVT_02 | Cycle2: Missing 'event_date' fixed. New bug found<br>Cycle3: Issue fixed       |
|                               |           | TC_DBC_EVT_04 | PUT    | /live-campus-hub/api/v1/events/name/Viramaya                                         | Update exist event           | {"event_title": "Medusa", "event_date": "2026-07-10", "event_time": "16:30:00", "location_name": "Car Park", "club_name": "Students Media Club"} | 204 No Content | Fail    | Fail | Pass | BUG_DBC_EVT_03 | Cycle2: Missing 'event_date' fixed. New bug found<br>Cycle3: Issue fixed       |
|                               |           | TC_DBC_EVT_05 | DELETE | /live-campus-hub/api/v1/events/event_50cf77b7-dafe-4a49-8702-65acc1feb01             | Delete exist event           | N/A                                                                                                                                              | 204 No Content | Fail    | Pass | Pass | BUG_DBC_EVT_04 | Cycle2: Issue fixed                                                            |
|                               | clubs     | TC_DBC_CLB_01 | GET    | /live-campus-hub/api/v1/clubs                                                        | Get all clubs                | N/A                                                                                                                                              | 200 OK         | Pass    | Pass | Pass | BUG_DBC_CLB_01 | Cycle2: Issue fixed                                                            |
|                               |           | TC_DBC_CLB_02 | POST   | /live-campus-hub/api/v1/clubs                                                        | Create new club              | {"club_name": "Rotaract Club", "status": true}                                                                                                   | 201 Created    | Fail    | Pass | Pass | BUG_DBC_CLB_02 | Cycle2: Issue fixed                                                            |
|                               |           | TC_DBC_CLB_03 | PUT    | /live-campus-hub/api/v1/clubs                                                        | Update exist club            | {"club_id": "club_3fca2fdd-af45-4234-896e-ac4addfda784", "club_name": "Badminton Club", "status": true}                                          | 204 No Content | Pass    | Pass | Pass | BUG_DBC_CLB_03 | Cycle2: Issue fixed                                                            |
|                               |           | TC_DBC_CLB_04 | PATCH  | /live-campus-hub/api/v1/clubs/club_5156a309-7b0a-4fe2-ba70-8f094569aeb3?status=false | Update only club status      | N/A                                                                                                                                              | 204 No Content | Fail    | Fail | Fail | BUG_DBC_CLB_04 | Used @PathVariable but no path is defined                                      |
|                               |           | TC_DBC_CLB_05 | DELETE | /live-campus-hub/api/v1/clubs/club_200f9273-723e-42f2-8dca-a5b4540a607b              | Delete exist club            | N/A                                                                                                                                              | 204 No Content | Pass    | Pass | Pass | -              | -                                                                              |
|                               |           | TC_DBC_STC_01 | GET    | /live-campus-hub/api/v1/students_club                                                | Get all student club records | N/A                                                                                                                                              | 200 OK         | Fail    | Pass | Pass | BUG_DBC_STC_01 | Cycle2: Missing 'joined_date', 'active_status' fixed.<br>Cycle3: New bug found |

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|  |  |  |               |      |                                     |                                     |                                                                                                                                                                                                                                                   |                 |      |      |   |                |                                                        |
|--|--|--|---------------|------|-------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|------|---|----------------|--------------------------------------------------------|
|  |  |  | TC_REG_STU_01 | POST | /live-campus-hub/api/v1/auth/signup | Register new student                | {<br>"user_id": "STU_005","password": "SahanLakmal123","<br>first_name": "Sahan","last_name": "Lakmal","email":<br>"sahan95@gmail.com","phone":"0712815499","<br>enrolled_Year": "2024","department_id": "D004","role":<br>"STUDENT"}<br>}        | 201 Created     | Pass | Pass | - | -              | Actual response is 200 OK<br>but should be 201 Created |
|  |  |  | TC_REG_STU_02 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing ID            | {<br>"user_id": "", "password": "sadanipwd#34","first_name":<br>"Sadani","last_name": "Wijethunga","email": "sadani.<br>wijethunga@gmail.com","phone":"0712362950","<br>enrolled_Year": "2026","department_id": "D005","role":<br>"STUDENT"}<br>} | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_01 |                                                        |
|  |  |  | TC_REG_STU_03 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing password      | {<br>"user_id": "STU_010","password": "", "first_name":<br>"Kumari","last_name": "Wijesinghe","email":<br>"kumariwije@gmail.com","phone":"0715432445","<br>enrolled_Year": "2024","department_id": "D005","role":<br>"STUDENT"}<br>}              | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_02 |                                                        |
|  |  |  | TC_REG_STU_04 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing first name    | {<br>"user_id": "STU_015","password": "sabina@234#B","<br>first_name": "", "last_name": "Hidalgo","email": "sabina.<br>h@gmail.com","phone":"0715472947","enrolled_Year":<br>"2021","department_id": "D004","role": "STUDENT"}<br>}               | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_03 |                                                        |
|  |  |  | TC_REG_STU_05 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing last name     | {<br>"user_id": "STU_016","password": "joshpwd123@B","<br>first_name": "Josh","last_name": "", "email":<br>"josh45@gmail.com","phone":"0782340952","<br>enrolled_Year": "2025","department_id": "D005","role":<br>"STUDENT"}<br>}                 | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_04 |                                                        |
|  |  |  | TC_REG_STU_06 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing email         | {<br>"user_id": "STU_014","password": "HeyoonJpwd#","<br>first_name": "Heyoon","last_name": "Jeong","email": "", "<br>phone":"0763410463","enrolled_Year": "2023","<br>department_id": "D004","role": "STUDENT"}<br>}                             | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_05 |                                                        |
|  |  |  | TC_REG_STU_07 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing phone number  | {<br>"user_id": "STU_017","password": "SinaD#123pwd","<br>first_name": "Sina","last_name": "Deinert","email":<br>"sinad23@gmail.com","phone": "", "enrolled_Year":<br>"2025","department_id": "D005","role": "STUDENT"}<br>}                      | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_06 |                                                        |
|  |  |  | TC_REG_STU_08 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing enrolled year | {<br>"user_id": "STU_018","password": "SinaD#123pwd","<br>first_name": "John","last_name": "Smith","email":<br>"john@gmail.com","phone":"07138295630","<br>enrolled_Year": "", "department_id": "D004","role":<br>"STUDENT"}<br>}                 | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_07 |                                                        |
|  |  |  | TC_REG_STU_09 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing department id | {<br>"user_id": "STU_019","password": "melvinF3pwd","<br>first_name": "Melvin","last_name": "Forbis","email":<br>"forbism@gmail.com","phone":"0724495738","<br>enrolled_Year": "2022","department_id": "", "role":<br>"STUDENT"}<br>}             | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_08 |                                                        |
|  |  |  | TC_REG_STU_10 | POST | /live-campus-hub/api/v1/auth/signup | Register with missing role          | {<br>"user_id": "STU_020","password":<br>"RickNovak12#pwd","first_name": "Rick","last_name":<br>"Novak","email": "novakrick@gmail.com","phone": "<br>0715420004","enrolled_Year": "2023","department_id":<br>"D005","role": ""}<br>}              | 400 Bad Request | Pass | Pass | - | -              |                                                        |
|  |  |  | TC_REG_STU_11 | POST | /live-campus-hub/api/v1/auth/signup | Register with already registered ID | {<br>"user_id": "STU_015","password": "joshpwd123@B","<br>first_name": "Josh","last_name": "Beauchamp","email":<br>"josh45@gmail.com","phone":"0782340952","<br>enrolled_Year": "2025","department_id": "D005","role":<br>"STUDENT"}<br>}         | 409 Conflict    | Pass | Pass | - | BUG_REG_STU_09 |                                                        |
|  |  |  | TC_REG_STU_12 | POST | /live-campus-hub/api/v1/auth/signup | Register with invalid email format  | {<br>"user_id": "STU_020","password": "neha12#pwd","<br>first_name": "Neha","last_name": "Sankalpana","email":<br>"sankalpanagmailcom","phone":"0763402993","<br>enrolled_Year": "2020","department_id": "D004","role":<br>"STUDENT"}<br>}        | 400 Bad Request | Fail | Fail | - | BUG_REG_STU_10 |                                                        |

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|  |  |  |               |      |                                    |                                        |                                                                              |                  |      |      |  |                |   |
|--|--|--|---------------|------|------------------------------------|----------------------------------------|------------------------------------------------------------------------------|------------------|------|------|--|----------------|---|
|  |  |  | TC_LOG_STU_01 | POST | /live-campus-hub/api/v1/auth/login | Login with valid username and password | {"username":"STU_014", "password": "dularik123pwd"}                          | 200 OK           | Pass | Pass |  | -              | - |
|  |  |  | TC_LOG_STU_02 | POST | /live-campus-hub/api/v1/auth/login | Login with invalid ID                  | {"username":"STU_000", "password": "dularik123pwd"}                          | 401 Unauthorized | Pass | Pass |  | BUG_LOG_STU_01 | - |
|  |  |  | TC_LOG_STU_03 | POST | /live-campus-hub/api/v1/auth/login | Login with invalid password            | {"username":"STU_014", "password": "password123"}                            | 401 Unauthorized | Pass | Pass |  | BUG_LOG_STU_01 | - |
|  |  |  | TC_LOG_STU_04 | POST | /live-campus-hub/api/v1/auth/login | Login with missing ID                  | {"username":"","password": "dularik123pwd"}                                  | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_02 | - |
|  |  |  | TC_LOG_STU_05 | POST | /live-campus-hub/api/v1/auth/login | Login with missing password            | {"username":"STU_014", "password": ""}                                       | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_02 | - |
|  |  |  | TC_LOG_STU_06 | POST | /live-campus-hub/api/v1/auth/login | Login with missing ID and password     | {"username":"","password": ""}                                               | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_02 | - |
|  |  |  | TC_LOG_STU_07 | POST | /live-campus-hub/api/v1/auth/login | SQL injection in ID field              | {"username":""," OR '1'=1", "password": "dularik123pwd"}                     | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_03 | - |
|  |  |  | TC_LOG_STU_08 | POST | /live-campus-hub/api/v1/auth/login | SQL injection in password field        | {"username":"STU_014", "password": "" OR '1'=1"}                             | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_03 | - |
|  |  |  | TC_LOG_STU_09 | POST | /live-campus-hub/api/v1/auth/login | Union injection in ID field            | {"username":""," UNION SELECT * FROM users --", "password": "dularik123pwd"} | 400 Bad Request  | Pass | Pass |  | BUG_LOG_STU_03 | - |
|  |  |  | TC_LOG_STF_01 | POST | /live-campus-hub/api/v1/auth/login | Login with valid username and password | {"username":"STF_001", "password": "hashen123#"}                             | 200 OK           | Pass | Pass |  | -              | - |
|  |  |  | TC_LOG_STF_02 | POST | /live-campus-hub/api/v1/auth/login | Login with invalid ID                  | {"username":"STF_000", "password": "hashen123#"}                             | 401 Unauthorized | Pass | Pass |  | BUG_REG_STF_01 | - |
|  |  |  | TC_LOG_STF_03 | POST | /live-campus-hub/api/v1/auth/login | Login with invalid password            | {"username":"STF_001", "password": "password123#"}                           | 401 Unauthorized | Pass | Pass |  | BUG_REG_STF_01 | - |
|  |  |  | TC_LOG_STF_04 | POST | /live-campus-hub/api/v1/auth/login | Login with missing ID                  | {"username":"","password": "hashen123#"}                                     | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_02 | - |
|  |  |  | TC_LOG_STF_05 | POST | /live-campus-hub/api/v1/auth/login | Login with missing password            | {"username":"STF_001", "password": ""}                                       | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_02 | - |
|  |  |  | TC_LOG_STF_06 | POST | /live-campus-hub/api/v1/auth/login | Login with missing ID and password     | {"username":"","password": ""}                                               | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_02 | - |
|  |  |  | TC_LOG_STF_07 | POST | /live-campus-hub/api/v1/auth/login | SQL injection in ID field              | {"username":""," OR '1'=1", "password": "hashen123#"}                        | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_03 | - |
|  |  |  | TC_LOG_STF_08 | POST | /live-campus-hub/api/v1/auth/login | SQL injection in password field        | {"username":"STF_001", "password": "" OR '1'=1"}                             | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_03 | - |
|  |  |  | TC_LOG_STF_09 | POST | /live-campus-hub/api/v1/auth/login | Union injection in ID field            | {"username":""," UNION SELECT * FROM users --", "password": "hashen123#"}    | 400 Bad Request  | Pass | Pass |  | BUG_REG_STF_03 | - |

|  | Feature    | Test ID      | Test Scenarios                    | Expected Result                                                        | Actual Result                                                | Cycle1 Status (Initial) | Cycle2 Status (Regression) | Cycle3 Status (Regression) | Bug ID        | Notes                                                                                   | Improvement ID |
|--|------------|--------------|-----------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------|----------------------------|----------------------------|---------------|-----------------------------------------------------------------------------------------|----------------|
|  |            |              |                                   |                                                                        | Pre Login Interface                                          |                         |                            |                            |               |                                                                                         |                |
|  | Branding   | TC_UI_PRE_01 | Campus name displays correctly    | "CINEC"                                                                | "Kenton"                                                     | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_01 | -                                                                                       | -              |
|  |            | TC_UI_PRE_02 | Color palette matches branding    | Navy blue + Sky blue                                                   | Navy blue + Gold                                             | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_02 | -                                                                                       | -              |
|  |            | TC_UI_PRE_03 | Consistent component design       | Consistent style across all components                                 | Consistent                                                   | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | -              |
|  | Login Page | TC_UI_PRE_04 | Login page loads correctly        | All elements visible                                                   | All visible                                                  | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | UI_IMP_01      |
|  |            | TC_UI_PRE_05 | Seperate login form exist         | Shows login form                                                       | Show login form                                              | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | -              |
|  |            | TC_UI_PRE_06 | Student ID field validation       | System blocks login and shows message like "Student ID required"       | System blocks login and shows "Please enter your Student ID" | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | UI_IMP_02      |
|  |            | TC_UI_PRE_07 | Password field validation         | System blocks login and shows messages like "Password required"        | login success with empty, invalid password                   | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_03 | Cycle2: System blocks login and shows "Please enter both your Student ID and Password." | UI_IMP_03      |
|  |            | TC_UI_PRE_08 | Login with empty fields           | System blocks login                                                    | System blocks login                                          | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | -              |
|  |            | TC_UI_PRE_09 | Password Minimum character length | Shows "Password must be at least X characters"                         | No minimum character length                                  | Fail                    | Fail                       | Deferred                   | BUG_PRE_UI_04 | Postponed due to time limitation                                                        | -              |
|  |            | TC_UI_PRE_10 | Password Masking                  | Password characters hidden                                             | Characters well hidden                                       | Pass                    | Pass                       | Pass                       | -             | -                                                                                       | -              |
|  |            | TC_UI_PRE_11 | Password visibility toggle        | Password visible with eye icon                                         | No visibility toggle                                         | Fail                    | Fail                       | Deferred                   | BUG_PRE_UI_05 | Postponed due to time limitation                                                        | -              |
|  |            | TC_UI_PRE_12 | Login successful feedback         | Shows "Login Successful"                                               | No feedback                                                  | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_06 | Cycle2: Shows "Login Successful"                                                        | UI_IMP_04      |
|  |            | TC_UI_PRE_13 | Back to home page button          | Include back buttons to redirect home page                             | "Back to homepage" link available in login form              | Pass                    | Fail                       | Pass                       | BUG_PRE_UI_07 | Cycle2: "Back to homepage" link removed                                                 | UI_IMP_05      |
|  |            | TC_UI_PRE_14 | Link with registration page       | Include a link to registration page                                    | No link between login page and registration page             | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_08 | Cycle2: Login page linked with registration page by "New here? Create new account"      | -              |
|  |            | TC_UI_PRE_15 | Seperate registration form exist  | Shows sign up form                                                     | Open same login form with Student ID and Password fields     | Fail                    | Pass                       | Pass                       | BUG_PRE_UI_09 | Cycle2: Seperate registration form exist                                                | UI_IMP_06      |
|  |            | TC_UI_PRE_16 | Full name field validation        | System blocks registration and shows message like "Full name required" | No validation                                                | Fail                    | Pass                       | Fail                       | BUG_PRE_UI_10 | Cycle2: Name required to register                                                       | UI_IMP_07      |
|  |            | TC_UI_PRE_17 | Student ID field validation       | System blocks login and shows message like "Student ID required"       | No validation                                                | Fail                    | Fail                       | Pass                       | BUG_PRE_UI_11 | -                                                                                       | UI_IMP_08      |
|  |            | TC_UI_PRE_18 | Email field validation            | System blocks login and shows message like "Email required"            | No validation                                                | Fail                    | Pass                       | Fail                       | BUG_PRE_UI_12 | Cycle2: Email required to register                                                      | UI_IMP_09      |

|  |              |               |                                   |                                                                                                  |                                                             |      |      |          |                |                                                                                                                                                                      |           |
|--|--------------|---------------|-----------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|------|----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|  | Sign Up Page | TC_UI_PRE_19  | Password field validation         | System blocks login and shows message like "Password required"                                   | No validation                                               | Fail | Fail | Pass     | BUG_PRE_UI_13  | -                                                                                                                                                                    | UI_IMP_10 |
|  |              | TC_UI_PRE_20  | Confirm Password field            | Include "Confirm Password" field                                                                 | No "Confirm Password" field                                 | Fail | Fail | Deferred | BUG_PRE_UI_14  | Postponed due to time limitation                                                                                                                                     | -         |
|  |              | TC_UI_PRE_21  | Register with empty fields        | System blocks registration                                                                       | System blocks registration                                  | Fail | Pass | Pass     | BUG_PRE_UI_15  | Cycle2: Registration blocks for empty fields                                                                                                                         | -         |
|  |              | TC_UI_PRE_22  | Password Minimum character length | Shows "Password must be at least X characters"                                                   | No minimum character length                                 | Fail | Pass | Deferred | BUG_PRE_UI_16  | Postponed due to time limitation                                                                                                                                     | -         |
|  |              | TC_UI_PRE_23  | Password Masking                  | Password characters hidden                                                                       | No password masking                                         | Fail | Pass | Pass     | BUG_PRE_UI_17  | Cycle2: Password masked properly                                                                                                                                     | -         |
|  |              | TC_UI_PRE_24  | Password visibility toggle        | Password visible with eye icon                                                                   | No visibility toggle                                        | Fail | Fail | Deferred | BUG_PRE_UI_18  | Postponed due to time limitation                                                                                                                                     | -         |
|  |              | TC_UI_PRE_25  | Registration Successful feedback  | Shows "Registration Successful"                                                                  | No feedback                                                 | Fail | Pass | Pass     | BUG_PRE_UI_19  | Cycle2: Shows "Registration Successful"                                                                                                                              | UI_IMP_11 |
|  |              | TC_UI_PRE_26  | Back to home page button          | Include back buttons to redirect home page                                                       | No back button                                              | Fail | Fail | Pass     | BUG_PRE_UI_20  | -                                                                                                                                                                    | UI_IMP_05 |
|  |              | TC_UI_PRE_27  | Link with login page              | Include a link to login page                                                                     | No link between registration page and login page            | Fail | Pass | Pass     | BUG_PRE_UI_21  | Cycle2: Registration page linked with login page by "Back to login"                                                                                                  | -         |
|  | Landing page | TC_UI_PRE_28  | Cards positioning                 | Cards properly positioned, no overlap                                                            | Cards partially overlapping the background                  | Fail | Fail | Pass     | BUG_PRE_UI_22  | -                                                                                                                                                                    | UI_IMP_12 |
|  | Responsive   | TC_UI_PRE_29  | Mobile view 375px                 | Layout adjusts properly                                                                          | Have some wrapping issues between login and Sign up buttons | Fail | Pass | Pass     | BUG_PRE_UI_23  | Cycle2: Wrapping issue fixed                                                                                                                                         | -         |
|  |              | TC_UI_PRE_30  | Tablet view 768px                 | Layout adjusts properly                                                                          | Layout adjusts                                              | Pass | Pass | Pass     | -              | -                                                                                                                                                                    | -         |
|  |              | TC_UI_PRE_31  | Desktop view 1920 * 1080 px       | Content centered properly                                                                        | Content centered                                            | Pass | Pass | Pass     | -              | -                                                                                                                                                                    | -         |
|  |              |               |                                   |                                                                                                  | Post Login Interface                                        |      |      |          |                |                                                                                                                                                                      |           |
|  | Dashboard    | TC_UI_POST_01 | Dashboard loads correctly         | Dashboard visible with all core features                                                         | Loads and visible                                           | Pass | Pass | Descoped | -              | Cycle3: Completely removed                                                                                                                                           | -         |
|  |              | TC_UI_POST_02 | User-friendly greeting            | Greet with user name                                                                             | Greet with student id                                       | Fail | Pass | Descoped | BUG_POST_UI_01 | Cycle2: Greets with name<br>Cycle3: Completely removed                                                                                                               | -         |
|  |              | TC_UI_POST_03 | Points balance displays           | Shows points balance                                                                             | Shows points balance                                        | Pass | Pass | Descoped | -              | Cycle3: Completely removed                                                                                                                                           | -         |
|  |              | TC_UI_POST_04 | All the core features included    | Main features including Transit, Events, Clubs, Checkin & Reward and Profile are clearly visible | Some features missing                                       | Fail | Fail | Descoped | BUG_POST_UI_02 | Cycle1: Missing Checkin, Rewards and Profile sections<br>Cycle2: Added Checkin, Rewards and Profile but removed Events and Clubs pages<br>Cycle3: Completely removed | -         |
|  |              | TC_UI_POST_05 | Quick actions                     | Quick action cards visible                                                                       | No quick actions                                            | Fail | Pass | Descoped | BUG_POST_UI_03 | Cycle2: Added quick actions<br>Cycle3: Completely removed                                                                                                            | -         |

|  |            |               |                                        |                                                                            |                                                                                              |      |      |          |                |                                                                           |           |
|--|------------|---------------|----------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------|------|----------|----------------|---------------------------------------------------------------------------|-----------|
|  |            | TC_UI_POST_06 | Informative status bar                 | Status bar with colors showing live updates                                | Visible                                                                                      | Pass | Pass | Descoped | -              | Cycle3: Completely removed                                                |           |
|  |            | TC_UI_POST_07 | Gamification elements                  | Include points balance, status indicators,                                 | All visible                                                                                  | Pass | Pass | Descoped | -              | Cycle3: Completely removed                                                |           |
|  | Navigation | TC_UI_POST_08 | Navigation bar visibility              | Shows links to main features                                               | Shows links                                                                                  | Pass | Pass | Pass     | -              | Cycle3: Removed top nav bar and added side nav bar                        | UI_IMP_13 |
|  |            | TC_UI_POST_09 | Navigational consistency               | Pages loads correctly                                                      | Pages loads                                                                                  | Pass | Pass | Pass     | -              | -                                                                         | -         |
|  |            | TC_UI_POST_10 | Active menu highlight                  | Active menu section has different style                                    | Active menu visible with a different style                                                   | Pass | Pass | Pass     | -              | -                                                                         | UI_IMP_14 |
|  |            | TC_UI_POST_11 | Top nav bar with bottom navbar         | Top navbar for desktop view and bottom navbar for mobile view              | Uses both                                                                                    | Pass | Fail | Fail     | BUG_POST_UI_04 | Cycle2: Removed top nav bar available bottom nav bar for both views       | UI_IMP_15 |
|  | Logout     | TC_UI_POST_12 | Logout functionality                   | Logs out correctly                                                         | Logs out and redirect to main page                                                           | Pass | Pass | Pass     | -              | Cycle3: Logout redirects to login page                                    | -         |
|  |            | TC_UI_POST_13 | Logout feedback                        | Display logout confirmation dialogue and logout successful message         | No visual feedback                                                                           | Fail | Pass | Fail     | BUG_POST_UI_05 | Cycle2: Added logout confirmation dialogue                                | UI_IMP_16 |
|  | Profile    | TC_UI_POST_14 | Profile picture upload                 | File picker opens to select an image                                       | Not working                                                                                  | Fail | Pass | Pass     | BUG_POST_UI_06 | Cycle2: File picker added and working properly                            | -         |
|  |            | TC_UI_POST_15 | Bio section exist                      | Bio section exist                                                          | No bio section exist                                                                         | Fail | Pass | Descoped | BUG_POST_UI_07 | Cycle2: Bio section exist<br>Cycle3: Colpetely removed                    | -         |
|  |            | TC_UI_POST_16 | Form fields display                    | Name, Email, Fields display                                                | No fields to display                                                                         | Fail | Pass | Pass     | BUG_POST_UI_08 | Cycle2: "Display Name" field available to type preferred name             | UI_IMP_17 |
|  | Transit    | TC_UI_POST_17 | Transit page loads with info           | Transit info visible                                                       | All visible                                                                                  | Pass | Pass | Pass     | -              | -                                                                         | -         |
|  | Events     | TC_UI_POST_18 | Events page loads with info            | Events info visible                                                        | All necessary info visible                                                                   | Pass | Fail | Pass     | BUG_POST_UI_09 | Cycle2: Events page removed completely                                    | -         |
|  | Clubs      | TC_UI_POST_19 | Clubs page loads with info             | Clubs info visible                                                         | Necessary info like club name, status, description available                                 | Pass | Fail | Pass     | BUG_POST_UI_10 | Cycle2: Clubs section removed completely                                  | -         |
|  |            | TC_UI_POST_20 | Clubs connector                        | Student able to discover clubs and join                                    | clubs connector missing                                                                      | Fail | Fail | Pass     | BUG_POST_UI_11 | Cycle1: No join option available<br>Cycle2: Clubs section removed         | -         |
|  | Check-in   | TC_UI_POST_21 | Check-in page loads                    | Checkin page loads properly                                                | No checkin page available                                                                    | Fail | Pass | Pass     | BUG_POST_UI_12 | Cycle2: Chckein page added                                                | -         |
|  |            | TC_UI_POST_22 | Relevant functionalities work properly | Open camera button available to scan QR codes at valid locations and works | No buttons                                                                                   | Fail | Fail | Fail     | BUG_POST_UI_13 | Cycle2: "Open Camera" button available but not working                    | -         |
|  | Rewards    | TC_UI_POST_23 | Redeemable rewards available           | Include a reward page with redeemable rewards                              | No reward page with redeemable rewards                                                       | Fail | Pass | Fail     | BUG_POST_UI_14 | Cycle2: Rewards page added with redeemable rewards                        | -         |
|  | Facility   | TC_UI_POST_24 | Display available facilities           | Show relevant info like availability status, open/close time with status   | No facility section showing facilities with availability status, open/close status and times | Fail | Fail | Pass     | BUG_POST_UI_15 | Cycle3: Displays facility name, capacity and status like "Quiet", "Empty" | -         |
|  |            | TC_UI_POST_25 | Mobile view 375px                      | Layout adjusts properly                                                    | Layout adjusts properly                                                                      | Pass | Pass | Fail     | BUG_POST_UI_16 | -                                                                         | -         |

[illegible]

|  | Test Type | Feature | Bug ID         | Bug Description                                                                                                      | Test ID       | Severity | Priority | Assign To         | Status | Steps to Reproduce                                                                                                                                                                               | Reported Date | Closed Date |
|--|-----------|---------|----------------|----------------------------------------------------------------------------------------------------------------------|---------------|----------|----------|-------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|
|  |           |         | BUG_DBC_DEP_01 | In departments table, 'department_name' field update to unreadable text                                              | TC_DBC_DEP_03 | Critical | High     | Backend Developer | Closed | 1. Send POST request to /departments.<br>2. Verify HTTP response status code as 201 Created<br>3. Navigate to departments table in database<br>4. Verify department_id update to unreadable text | 12/04/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_STD_01 | Student entity class and student table have 'enrolled_year' in different data types                                  | TC_DBC_STD_01 | High     | High     | Backend Developer | Closed | 1. Send GET request to /students<br>2. Verify HTTP response as 500 Internal Server Error                                                                                                         | 31/03/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_STD_02 | Authentication error                                                                                                 | TC_DBC_STD_02 | High     | High     | Backend Developer | Closed | 1. Send GET request to /students/me<br>2. Verify HTTP response as 500 Internal Server Error                                                                                                      | 12/04/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_STD_03 | Due to data type mismatch of 'enrolled_year' code tries to store NULL but DB rejects because of Not Null constraints | TC_DBC_STD_03 | Critical | High     | Backend Developer | Closed | 1. Send POST request to /students<br>2. Verify HTTP response as 500 Internal Server Error                                                                                                        | 31/03/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_STD_04 | Authentication error                                                                                                 | TC_DBC_STD_05 | High     | High     | Backend Developer | Closed | 1. Send PUT request to /students/me<br>2. Verify HTTP response as 500 Internal Server Error                                                                                                      | 12/04/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_STD_05 | Student table and student entity class have 'enrolled_year' in different data types. This cause for Delete requests  | TC_DBC_STD_06 | High     | High     | Backend Developer | Closed | 1. Send DELETE request to /students/{id}<br>2. Verify HTTP response as 500 Internal Server error                                                                                                 | 31/03/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_RWD_01 | In rewards table, 'reward_id' field update to unreadable text                                                        | TC_DBC_RWD_03 | Critical | High     | Backend Developer | Closed | 1. Send POST request to /rewards<br>2. Verify HTTP response status code as 201 Created<br>3. Navigate to rewards table in database<br>4. Verify reward_id update to unreadable text              | 12/04/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_LOC_01 | In locations table, 'location_id' field update to unreadable text                                                    | TC_DBC_LOC_05 | Critical | High     | Backend Developer | Closed | 1. Send POST request to /locations<br>2. Verify HTTP response as 201 created<br>3. Navigate to locations table in database<br>4. Verify location_id update to unreadable text                    | 12/04/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_EVT_01 | Failed to retrieve 'event_date' in event table due to missing 'event_date' attribute in Event entity class           | TC_DBC_EVT_01 | High     | High     | Backend Developer | Closed | 1. Send GET request to /events<br>2. Verify status code as 500 Internal Server error                                                                                                             | 31/03/2026    | 25/05/2026  |
|  |           |         | BUG_DBC_EVT_02 | Failed to create records in event table due to missing 'event_date' attribute in Event entity class                  | TC_DBC_EVT_03 | High     | High     | Backend Developer | Closed | 1. Send POST request to /events<br>2. Verify status code as 500 Internal Server error                                                                                                            | 31/03/2026    | 25/05/2026  |
|  |           |         |                | 'club_id', 'event_title', 'location_id' store as null                                                                |               | High     | High     |                   | Closed | 1. Send POST request to /events<br>2. Verify status code as 201 Created<br>3. Check database 'club_id', 'event_title', 'location_id' columns                                                     | 03/05/2026    | 25/05/2026  |
|  |           |         |                | Failed to update records in event table due to missing 'event_date' attribute in Event entity class                  |               | High     | High     |                   | Closed | 1. Send PUT request to /events<br>2. Verify status code as 500 Internal Server error                                                                                                             | 31/03/2026    | 25/05/2026  |

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|-------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|--------|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
| Database & Backend Connection | BUG_DBC_EVT_03 | 'club_id', 'event_title', 'location_id' update as null                                                                                                        | TC_DBC_EVT_04 | High     | High   | Backend Developer | Closed | 1. Send POST request to /events<br>2. Verify status code as 204 No Content<br>3. Check database 'club_id', 'event_title', 'location_id' columns                        | 03/05/2026 | 25/05/2026 |
|                               | BUG_DBC_EVT_04 | Failed to delete records in event table due to missing 'event_date' attribute in Event entity class                                                           | TC_DBC_EVT_05 | High     | High   | Backend Developer | Closed | 1. Send DELETE request to /events<br>2. Verify status code as 500 Internal Server error                                                                                | 31/03/2026 | 25/05/2026 |
|                               | BUG_DBC_CLB_01 | Retrieved 'club_id' as null even though data available in database. Club table doesn't include 'student_id' as a FK yet code tries to retrieve a 'student_id' | TC_DBC_CLB_01 | High     | High   | Backend Developer | Closed | 1. Send GET request to /clubs<br>2. Verify HTTP response status code as 200 OK<br>3. Verify club_id retrieve as NULL                                                   | 08/04/2026 | 25/05/2026 |
|                               | BUG_DBC_CLB_02 | Couldn't create a new club record due to unnecessary 'student_id' attribute                                                                                   | TC_DBC_CLB_02 | Medium   | Medium | Backend Developer | Closed | 1. Send POST request to /clubs<br>2. Verify HTTP response status code as 500 Internal Server error                                                                     | 12/04/2026 | 25/05/2026 |
|                               | BUG_DBC_CLB_03 | In clubs table, 'club_name' update as NULL                                                                                                                    | TC_DBC_CLB_03 | High     | High   | Backend Developer | Closed | 1. Send PUT request to /clubs<br>2. Verify HTTP response status code as 204 No Content<br>3. Navigate to clubs table in database<br>4. Verify club_name update as NULL | 12/04/2026 | 25/05/2026 |
|                               | BUG_DBC_CLB_04 | Couldn't update 'status' in clubs table because path variable for id in PATCH mapping not defined                                                             | TC_DBC_CLB_04 | High     | High   | Backend Developer | Open   | 1. Send PATCH request to /clubs<br>2. Check club controller<br>3. Observe error in console                                                                             | 12/04/2026 |            |
|                               | BUG_DBC_STC_01 | "joined_date" and "active_status" attributes missing in StudentClub entity class                                                                              | TC_DBC_STC_01 | Medium   | Medium | Backend Developer | Closed | 1. Send GET request to /students_club<br>2. Verify HTTP response as 500 Internal Server Error                                                                          | 31/03/2026 | 25/05/2026 |
|                               |                | "joined_date" returned as null                                                                                                                                |               | High     | High   |                   | Open   | 1. Send GET request to /students_club<br>2. Observe joined date returned as null                                                                                       | 21/05/2026 |            |
|                               | BUG_DBC_STC_02 | Authentication error                                                                                                                                          | TC_DBC_STC_02 | High     | High   | Backend Developer | Closed | 1. Send POST request to /students_club/join<br>2. Verify HTTP response as 500 Internal Server Error                                                                    | 12/04/2026 | 25/05/2026 |
|                               |                | "joined_date" store as null                                                                                                                                   |               | High     | High   |                   | Open   | 1. Send POST request to /students_club/join<br>2. Verify HTTP response as 200 OK<br>3. Check database record                                                           | 11/05/2026 |            |
|                               | BUG_DBC_STC_03 | Authentication error                                                                                                                                          | TC_DBC_STC_03 | High     | High   | Backend Developer | Closed | 1. Send PUT request to /students_club/C002<br>2. Verify HTTP response as 500 Internal Server Error                                                                     | 12/04/2026 | 25/05/2026 |
|                               | BUG_DBC_STC_04 | Authentication error                                                                                                                                          | TC_DBC_STC_04 | High     | High   | Backend Developer | Closed | 1. Send DELETE request to /students_club/leave<br>2. Verify HTTP response as 500 Internal Server Error                                                                 | 12/04/2026 | 25/05/2026 |
|                               | BUG_DBC_BUS_01 | In database, bus table missing 'bus_number' and 'status' fields                                                                                               | TC_DBC_BUS_01 | Critical | High   | Database Manager  | Closed | 1. Send GET request to /bus/getall<br>2. Verify HTTP response as 200 OK<br>3. Compare with Bus entity class                                                            | 31/03/2026 | 25/05/2026 |



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|----------------|-------------------------------------------------------------------------------------|---------------|----------|------|-------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|
| BUG_REG_STU_03 | Student signup accepts and creates record with empty first name                     | TC_REG_STU_04 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "first_Name".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty first name       | 02/05/2026 |  |
| BUG_REG_STU_04 | Student signup accepts and creates record with empty last name                      | TC_REG_STU_05 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "last_Name".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty last name         | 02/05/2026 |  |
| BUG_REG_STU_05 | Student signup accepts and creates record with empty email                          | TC_REG_STU_06 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "email".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty email                 | 02/05/2026 |  |
| BUG_REG_STU_06 | Student signup accepts and creates record with empty phone number                   | TC_REG_STU_07 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "phone".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty phone                 | 02/05/2026 |  |
| BUG_REG_STU_07 | Student signup accepts and creates record with empty enrolled year                  | TC_REG_STU_08 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "enrolled_Year".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty enrolled year | 02/05/2026 |  |
| BUG_REG_STU_08 | Student signup accepts and creates record with empty department Id                  | TC_REG_STU_09 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "department_id".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty dep Id        | 02/05/2026 |  |
| BUG_REG_STU_09 | Register with already registered ID returns 400 Bad Request instead of 409 Conflict | TC_REG_STU_11 | Low      | Low  | Backend Developer | Open | 1. Register new user with Id<br>2. Send POST request to /signup with same Id<br>2. Observe response 400 Bad Request instead of 409 Conflict                          | 02/05/2026 |  |
| BUG_REG_STU_10 | Student signup accepts invalid email format                                         | TC_REG_STU_12 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with invalid email format<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with invalid email     | 02/05/2026 |  |
| BUG_REG_STU_11 | Student signup accepts short weak passwords                                         | TC_REG_STU_13 | High     | High | Backend Developer | Open | 1. Send POST request to /signup with "password":"123"<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with weak password         | 02/05/2026 |  |
| BUG_REG_STF_01 | Staff signup accepts and creates record with empty Id                               | TC_REG_STF_02 | Critical | High | Backend Developer | Open | 1. Send POST request to /signup with "userId".""<br>2. Observe response as 201 Created or 200 OK<br>3. Check database record created with empty user Id              | 02/05/2026 |  |



|  |  |                     |                |                                                                                          |               |        |        |                    |        |                                                                                                                              |            |            |
|--|--|---------------------|----------------|------------------------------------------------------------------------------------------|---------------|--------|--------|--------------------|--------|------------------------------------------------------------------------------------------------------------------------------|------------|------------|
|  |  | Login functionality | BUG_REG_STF_01 | Staff login with invalid details return 403 Forbidden instead of 401 Unauthorized        | TC_LOG_STF_02 | Low    | Low    | Backend Developer  | Open   | 1. Send POST to /login with non-existent user Id or invalid password<br>2. Observe 403 Forbidden instead of 401 unauthorized | 02/05/2026 |            |
|  |  |                     |                |                                                                                          | TC_LOG_STF_03 |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     | BUG_REG_STF_02 | Staff login with empty details return 403 Forbidden instead of 400 Bad Request           | TC_LOG_STF_04 | Medium | Medium | Backend Developer  | Open   | 1. Send POST to /login with empty user Id and password<br>2. Observe 403 Forbidden instead of 400 Bad Request                | 02/05/2026 |            |
|  |  |                     |                |                                                                                          | TC_LOG_STF_05 |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     |                |                                                                                          | TC_LOG_STF_06 |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     | BUG_REG_STF_03 | Staff login with SQL injection in fields return 403 Forbidden instead of 400 Bad Request | TC_LOG_STF_07 | Low    | Low    | Backend Developer  | Open   | 1. Send POST to /login with ' OR '1'='1' in Id or password field<br>2. Observe 403 Forbidden instead of 400 Bad Request      | 02/05/2026 |            |
|  |  |                     |                |                                                                                          | TC_LOG_STF_08 |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     |                |                                                                                          | TC_LOG_STF_09 |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     |                |                                                                                          |               |        |        |                    |        |                                                                                                                              |            |            |
|  |  |                     | BUG_PRE_UI_01  | App uses "Kenton" instead of "CINEC"                                                     | TC_UI_PRE_01  | High   | High   | Frontend Developer | Closed | 1. Open main dashboard<br>2. Check campus name                                                                               | 03/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_02  | Color palette does not match the branding                                                | TC_UI_PRE_02  | High   | High   | Frontend Developer | Closed | 1. Open application<br>2. Check color palette used<br>3. Compare with Navy blue and Sky blue                                 | 03/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_03  | Password field not validated. Only Student ID required to login                          | TC_UI_PRE_07  | High   | High   | Frontend Developer | Closed | 1. Open login page<br>2. navigate to login page<br>3. Fill only Student ID field<br>4. Click login                           | 03/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_04  | Minimum character length is not displayed for password field in login form               | TC_UI_PRE_09  | Medium | Medium | Frontend Developer | Open   | 1. Open login page<br>2. Click on password field<br>3. Check if text shows minimum length                                    | 17/04/2026 |            |
|  |  |                     | BUG_PRE_UI_05  | Password field does not include a visibility toggle                                      | TC_UI_PRE_11  | Low    | Medium | Frontend Developer | Open   | 1. Open login page<br>2. Check password field<br>3. Check for eye icon to show/hide password                                 | 17/04/2026 |            |
|  |  |                     | BUG_PRE_UI_06  | "Login Successful" feedback is not displayed after login successfully                    | TC_UI_PRE_12  | Medium | Medium | Frontend Developer | Closed | 1. Open login page<br>2. Enter valid details<br>3. Click login<br>4. Verify if message appears                               | 17/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_07  | "Back to homepage" link removed in login form                                            | TC_UI_PRE_13  | Low    | Medium | Frontend Developer | Closed | 1. Open login page<br>2. Check "Back to homepage"<br>3. Verify if its available                                              | 17/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_08  | Login page does not include a link to registration page                                  | TC_UI_PRE_14  | High   | High   | Frontend Developer | Closed | 1. Open login page<br>2. Check if "register here" link available                                                             | 17/04/2026 | 25/05/2026 |
|  |  |                     | BUG_PRE_UI_09  | No seperate registration form to link Sign Up button                                     | TC_UI_PRE_15  | High   | High   | Frontend Developer | Closed | 1. Open application<br>2. Check if Sign Up button opens seperate registration page                                           | 17/04/2026 | 25/05/2026 |



|  |  |                |                                                                     |               |          |        |                    |        |                                                                                                                                              |            |            |
|--|--|----------------|---------------------------------------------------------------------|---------------|----------|--------|--------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|
|  |  | BUG_POST_UI_01 | Dashboard greets user with user id instead of user name             | TC_UI_POST_02 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. View dashboard welcome message<br>3. Check if shows student id or name                                           | 03/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_02 | Some main features missing                                          | TC_UI_POST_04 | Critical | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Look for Check-in, Rewards, Transit section                                                                      | 03/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_03 | No quick actions available                                          | TC_UI_POST_05 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Scroll to bottom of dashboard<br>3. Check for quick actions                                                      | 03/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_04 | Bottom navbar for both desktop and mobile view                      | TC_UI_POST_11 | Low      | Low    | Frontend Developer | Closed | 1. Login successfully<br>2. Check bottom of page for navigation bar                                                                          | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_05 | No logout confirmation dialogue and logout successful message       | TC_UI_POST_13 | Medium   | Medium | Frontend Developer | Open   | 1. Login successfully<br>2. Click logout button<br>3. Check if confirmation dialog appears<br>4. Check if logout success message appears     | 03/04/2026 |            |
|  |  | BUG_POST_UI_06 | No file picker open to select an image to upload as profile picture | TC_UI_POST_14 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Navigate to profile section<br>3. Look for profile picture upload button<br>4. Click to see if file picker opens | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_07 | No bio section to add a bio                                         | TC_UI_POST_15 | Medium   | Medium | Frontend Developer | Closed | 1. Login successfully<br>2. Go to profile page<br>3. Look for bio text area or field                                                         | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_08 | No fields like name, email to display in profile section            | TC_UI_POST_16 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Go to profile page<br>3. Check if name, email fields exist                                                       | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_09 | Events section removed                                              | TC_UI_POST_18 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Check navigation menu<br>3. Look for events page or link                                                         | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_10 | Clubs section removed                                               | TC_UI_POST_19 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Check navigation menu<br>3. Look for clubs page or link                                                          | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_11 | Users don't have the ability to discover and join clubs             | TC_UI_POST_20 | High     | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Navigate to clubs section<br>3. Try to view clubs list<br>4. Try to join a club                                  | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_12 | No checkin page available                                           | TC_UI_POST_21 | Critical | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Look for check-in page or section<br>3. Try to perform check-in                                                  | 17/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_13 | No relevant buttons available to perform checkin                    | TC_UI_POST_22 | Critical | High   | Frontend Developer | Closed | 1. Login successfully<br>2. Go to check-in page if exist<br>3. Look for check-in button or action                                            | 03/04/2026 | 25/05/2026 |
|  |  | BUG_POST_UI_14 | No reward page available with redeemable rewards                    | TC_UI_POST_23 | Critical | High   | Frontend Developer | Open   | 1. Login successfully<br>2. Look for rewards page<br>3. Check if redeemable rewards are shown                                                | 03/04/2026 |            |

|  |  |  |                |                                                                                                                    |               |          |      |                    |        |                                                                                                                             |            |  |
|--|--|--|----------------|--------------------------------------------------------------------------------------------------------------------|---------------|----------|------|--------------------|--------|-----------------------------------------------------------------------------------------------------------------------------|------------|--|
|  |  |  | BUG_POST_UI_15 | No seperate facility section to show available facilities' open/close time, open/close status, availability status | TC_UI_POST_24 | Critical | High | Frontend Developer | Closed | 1. Login successfully<br>2. Look for facilities section<br>3. Check if open. close time and status shown                    | 17/04/2026 |  |
|  |  |  | BUG_POST_UI_16 | Only sidebar properly visible content not visible                                                                  | TC_UI_POST_25 | High     | High | Frontend Developer | Open   | 1. Login successfully<br>2. Open dev tools and set responsive size as 375px<br>3. Observe responsiveness of the application | 23/05/2026 |  |

|  | Source     | Improvement ID | Sugestion                                                              | Current Behavior                                                                                                | Proposed Solution                                                                                                        | Priority | Status      | Reported Date | Reported By |
|--|------------|----------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|-------------|---------------|-------------|
|  | UI Testing | UI_IMP_01      | Fill background space in login page                                    | Login page only have login form.                                                                                | Use background image, gradient color or animation                                                                        | Medium   | Implemented | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_02      | Seperate validation message for Student ID field in login page         | When login only with password, displays common message like "Please enter both your Student ID and Password."   | Display seperate message like "Student ID required" or "Please enter your Student ID"                                    | Medium   | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_03      | Seperate validation message for Password field in login page           | When login only with Student ID, displays common message like "Please enter both your Student ID and Password." | Display seperate message like "Password required" or "Please enter your Password"                                        | Medium   | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_04      | Display "Login Successful" feedback after login successfully           | "Login Successful" message appear in top-right corner                                                           | Display the message in center to improve visibility                                                                      | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_05      | Include a back button to redirect to home page                         | Only browser back button available                                                                              | Add a "Back to Home" icon to improve navigation flexibility                                                              | Medium   | Implemented | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_06      | Name field improvement                                                 | Registration form only have full name field but database use seperate first name and last name columns          | Remove Full Name field and add First Name and Last Name fields                                                           | High     | Implemented | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_07      | Seperate validation message for Full Name field in registration page   | When register with only other fields, displays common message like "Please enter a valid name and email"        | Display seperate message like "Full Name required" or "Please enter your Full Name"                                      | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_08      | Seperate validation message for Student ID field in registration page  | When register with only other fields, displays common message like "Please enter a valid name and email"        | Display seperate message like "Student ID required" or "Please enter your Student ID"                                    | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_09      | Seperate validation message for Email field in registration page       | When register with only other fields, displays common message like "Please enter a valid name and email"        | Display seperate message like "Email required" or "Please enter your Email"                                              | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_10      | Seperate validation message for Password field in registration page    | When register with only other fields, displays common message like "Please enter a valid name and email"        | Display seperate message like "Password required" or "Please enter your Password"                                        | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_11      | Display "Registration Successful" feedback after register successfully | "Welcome to CINEC Hub, [Name]" message appear in top-right corner                                               | Display the message in center to improve visibility                                                                      | High     | Proposed    | 17/04/2026    | QA Tester   |
|  |            | UI_IMP_12      | Landing page lacks strong visual enagement for first time users        | Displays welcome message with brief description, navigation bar, login and sign up buttons                      | Enhance the hero section by adding CTA buttons, visula elements (illustration or image) and highlight main core features | High     | Proposed    | 17/04/2026    | QA Tester   |

|  |  |           |                                                                  |                                                                                               |                                                                                                                                                 |        |             |            |           |
|--|--|-----------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|------------|-----------|
|  |  | UI_IMP_13 | Emojis in navbar look informal                                   | Navbar use emojis to represent menu items                                                     | Replace emojis with a consistent icon set to improve usability and professionalism                                                              | Medium | Implemented | 17/04/2026 | QA Tester |
|  |  | UI_IMP_14 | Underline is not enough to notice active menu section            | Active menu item use blue underline                                                           | Enhance by adding a background color or different text color for better visibility                                                              | Medium | Implemented | 17/04/2026 | QA Tester |
|  |  | UI_IMP_15 | Displaying multiple menu items reduce clarity                    | Same bottom navbar is used for both desktop and mobile view by containing multiple menu items | Use a top navbar for desktop view and keep bottom navbar only for core features on mobile view and move additional links using a hamburger menu | High   | Implemented | 17/04/2026 | QA Tester |
|  |  | UI_IMP_16 | Display "Logout Successfully" feedback after logout successfully | "Logout Successfully" message appear in top-right corner                                      | Display the message in center to improve visibility                                                                                             | High   | Proposed    | 17/04/2026 | QA Tester |
|  |  | UI_IMP_17 | Display Name, Email fields in profile section                    | Available only Name field                                                                     | Add an Email field to retrieve and display user email                                                                                           | Medium | Implemented | 17/04/2026 | QA Tester |
|  |  | UI_IMP_18 | Use consistent font size for texts in buttons                    | Texts in buttons not cleraly visible                                                          | Increase font size in all buttons for better visibility                                                                                         | High   | Implemented | 17/04/2026 | QA Tester |
|  |  |           |                                                                  |                                                                                               |                                                                                                                                                 |        |             |            |           |
|  |  |           |                                                                  |                                                                                               |                                                                                                                                                 |        |             |            |           |